1 Basics of Semiconductor Physics

INTRODUCTION

- As this chapter is mainly an introduction to solid state devices, it is best to avoid referencing concepts like atomic theory, quantum physics, or electron models.
- The behaviour of solid state devices, on the other hand, is intrinsically linked to these concepts. The movement of an electron in a semiconductor can be studied after the knowledge of an electron and its interaction with the crystal lattice.
- We will start with an overview of the fundamental atomic properties of matter.
- Atomic energy levels in the crystal are spread out into energy bands. The energy band diagram can help us to differentiate between an insulator, a semiconductor, and a metal.

Definitions

Semiconductors:
These are the materials which have conductivity in between conductors (metals) and insulators. Semiconductors are of two types: one is pure (elemental) semiconductor such as silicon (Si) and germanium (Ge), other is compound semiconductor such as gallium arsenide (GaAs) or gallium phosphide (GaP).

NATURE OF ATOM

- In order to explain many phenomena associated with conduction in gases, metal and semiconductors and the emission of electrons from the surface of the metal, it is necessary to assume that the atom has loosely bound electrons which can be taken away from it.
- Rutherford found that an atom is made up of a positive-charge nucleus that contains almost all of the atom's mass, and hence, it will remain substantially immobile.
- Negatively charged electrons surround the central positive core.
- The force of attraction between the electron and proton follows Coulomb's law.
- Let us consider a hydrogen atom for atomic model. The atom has a proton and an electron.
  Proton → +ve charge
  Electron → –ve charge

Grey Matter Alert!

"The birth history of semiconductors can be traced back to the invention of the rectifier (AC-DC converter) in 1874. Decades later, Bardeen and Brattain at Bell Laboratories in the US invented the point-contact transistor in 1947, and Shockley invented the junction transistor in 1948."

- The attraction force between the nucleus and the electron is given by:

$$F = \frac{1}{4\pi \in_0 r^2} \cdot q^2 \quad \text{...1.1)}$$

where, q = electronic charge (in Coulombs)
r = distance between the two particles (in meters)
$\in_0$ = permittivity of free space
F = force (in Newton)

- According to Newton's second law of motion, "The net force must be equal to the product of an electronic mass (m) and acceleration $\left(\frac{v^2}{r}\right)$ towards the nucleus, where electron's speed in the circular path is denoted by 'v' in m/sec."

Then, $\frac{1}{4\pi \in_0 r^2} \cdot q^2 = \frac{mv^2}{r}$ ...(1.2)

- Potential energy (P.E.) of an electron at a distance 'r' from the nucleus is:

$$P.E. = \frac{-1}{4\pi \in_0 r} \cdot q^2 \quad \text{...(1.3)}$$

- Kinetic energy of the electron is:

$$K.E. = \frac{1}{2}mv^2 \quad \text{...1.4)}$$

- Total mechanical energy:

$$W = \frac{1}{2}mv^2 - \frac{1}{4\pi \in_0 r} \cdot q^2 \quad \text{...(1.5)}$$

$$W = -\frac{1}{8\pi \in_0 r} \cdot q^2 \quad \text{...(1.6)}$$

This equation shows that the total energy of the electron is always negative.

Grey Matter Alert!

An electron cannot stay at every energy state. It can last only at those energy states where its angular momentum is an integral multiple of $\frac{h}{2\pi}$ i.e. $mvr = \frac{nh}{2\pi}$

Previous Years' Question

The longest wavelength that can be absorbed by silicon, which has the band gap of 1.12 eV is 1.1 μm. If the longest wavelength that can be absorbed by an another material is 0.87 μm, then the band gap of this material is:

a) 1.452 eV **b)** 0.886 eV
c) 1.425 eV **d)** 0.706 eV

ATOMIC ENERGY LEVELS

- Stationary state is defined by the condition that the electron's angular momentum in this state is quantized and must be an integral multiple of $\frac{h}{2\pi}$.

  Thus, $mvr = \frac{nh}{2\pi}$ ...(1.7)

- Combining equations 1.2) and 1.7), we obtain the radii of the stationary state, and from equation 1.6), each state's energy level in Joules is determined as:

$$W_n = -\frac{mq^4}{8h^2 \epsilon_0^2} \cdot \frac{1}{n^2} \quad ...(1.8)$$

- For each integral value of 'n' in equation 1.8), a horizontal line is drawn. These lines are vertical under numerical values calculated from equation 1.8).
- The pictorial representation of equation 1.8) is called an energy level diagram. The energy level diagram for hydrogen is indicated in figure 1.1).

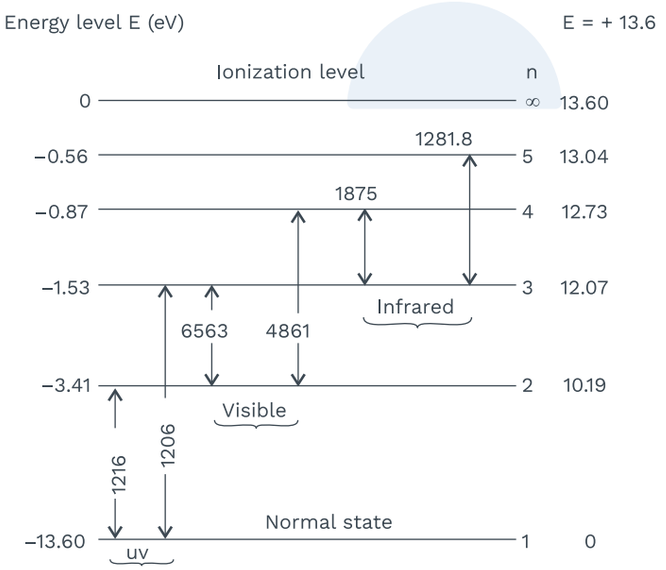


Fig. 1.1 The Ionization Level of Hydrogen

- The number to the left of each line represents the energy level's in electron volts (eV).

- The value of n is the number immediate to the right of a line.
- The relation between the difference of two energy states and wavelength λ of emitted radiation is:

$$\lambda(\text{cm}) = \frac{12400}{(E_2 - E_1)\,\text{eV}} \quad \ldots(1.9)$$

$$\text{Or, } E_2 - E_1 = \frac{1.24}{\lambda\,(\mu\text{m})}\,\text{eV} \quad \ldots(1.10)$$

$$\because \quad E_2 - E_1 = E_G$$

$$\because \quad E_G = \frac{1.24}{\lambda\,(\mu\text{m})}\,\text{eV} \quad \ldots(1.11)$$

ENERGY BANDS/BAND THEORY FOR MATERIAL CLASSIFICATION



Fig. 1.2 Potential Well

- Atom is represented in the form of potential well as shown in figure 1.2.
- In figure 1.2, the nucleus is the reference and left possible energies are represented as potential energy levels or energy levels.
- Allowed potential energy must satisfy the following condition:

  Angular momentum of the electron:

$$mvr = \frac{nh}{2\pi}$$

where

h = Planck's constant

n = 1, 2, 3, ...

- Silicon is an atom having 14 electrons.

Electronic configuration of Si = $1s^2$, $2s^2$, $2p^6$, $3s^2$, $3p^2$

Definitions

Potential well: A region in a force field, particularly the region in which an atomic nucleus is located, where the potential is significantly lower than at points immediately outside it, so that a particle in it is likely to remain there unless it gains a relatively a large amount of energy.

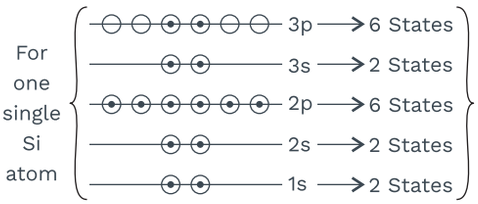


Fig. 1.3 Energy States of Single Si Atom

These energy levels are also called discrete energy levels because these energies are discrete in nature.

- But during the material formation, trillion and trillions of Avogadro's number of atoms will be there.

For N Silicon atoms:

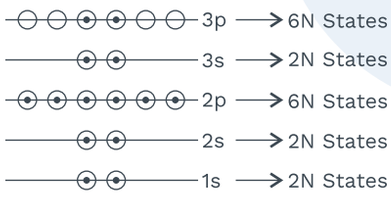


Fig. 1.4 Energy States of N Si Atom

- During the material formation, when two atoms come close to each other, their discrete energy levels will start interacting with each other and will start splitting into bands.

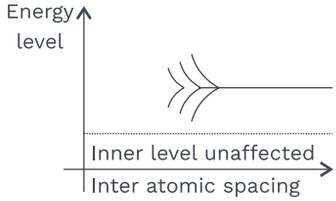


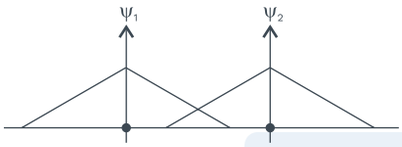
Fig. 1.5 Band Formation from Discrete Energy Levels

- When two silicon atoms are brought close together, the outermost shell or valence shell, (n=3) in which two 3s and two 3p electrons interact to form four hybridised" sp^3 electrons.

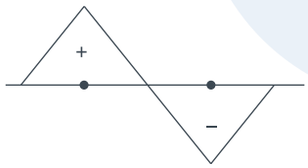
Rack Your Brain

Why inner energy levels of atoms are unaffected during the formation of material?

- We find that the composite two-electron wave functions are Linear Combinations of individual Atomic Orbitals (LCAO) by solving the Schrödinger equation for such an interacting system.
- The antibonding orbital is odd or antisymmetric combination, while the bonding orbitals are the even or symmetric combination.



Atomic Orbitals
Fig. 1.6 a)



Anti-bonding Orbitals
Fig. 1.6 b)



Bonding Orbitals
Fig. 1.6 c)

Fig. 1.6 Representation of Atomic Orbitals, Antibonding Orbitals, Bonding Orbitals.

- Many atoms are brought together into a solid, forming continuous bands of energies from the split energy levels.
- These energy levels are divided into bands as the interatomic distance decreases, starting with the outer shell (n=3). As the “3s” and “3p” bond grows, they merge into a single band of varying energy levels.
- This band of "3s-3p" level consists of 8N available states.

Rack Your Brain

Why are antibonding orbitals antisymmetric?

Grey Matter Alert!

A band gap is the energy range where no electronic state can exist.

- There is a split in the band separated by an energy band gap as the distance between the atoms approaches the equilibrium interatomic spacing of Si.
- The upper band contains 4N states, which are empty and is known as the conduction band.
- The lower band contains 4N states, which are filled completely and is known as the valence band.
- In a perfect crystal, there is no energy state in the energy gap and, therefore, it is also called as forbidden energy gap.

Definitions

Band gap: The energy difference between the valence and conduction bands of a solid material (such as an insulator or semiconductor) that consists of the range of energy values forbidden to electrons in the material.

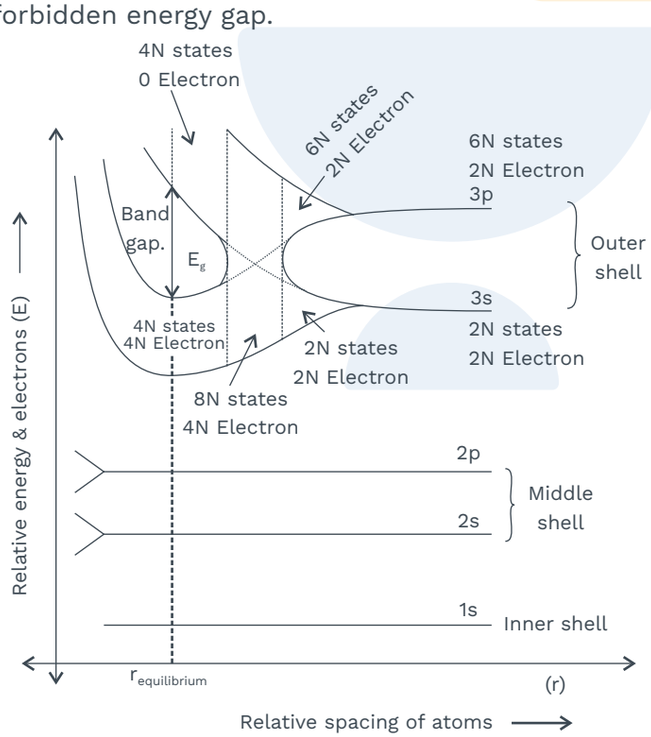


Fig. 1.7 Energy level in Si

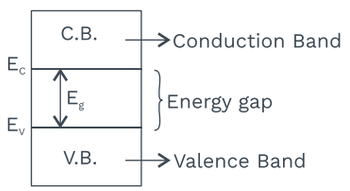


Fig. 1.8 Energy Band Diagram

Where

E_C = Lowest energy level of the conduction band

E_V = Highest energy level of the valence band

$E_g = E_C - E_V$ = energy gap

$r_{equilibrium}$ = Equilibrium interatomic spacing

CLASSIFICATION OF MATERIALS BASED ON ENERGY BAND DIAGRAM

- Every solid has a distinct energy band structure, and therefore, every solid has different electrical properties.
- On the basis of the energy band diagram, materials can be classified into three categories:
  1) Metals (Conductors)
  2) Insulators
  3) Semiconductors

1) Metals:

- In metals, conduction band and valence band overlaps each other, as shown in figure 1.9.

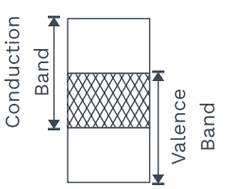


Fig. 1.9 Representation of Energy Bands in Metals

- There is no forbidden energy gap between the conduction band and the valence band.
- There is an overlap in the conduction band and valence band and hence, the filled states and the empty states overlap each other, which allows the electrons to move freely.
- Metal contains a large number of free electrons without any supply of additional energy, and that is why metal works as a good conductor.

For example, Cu, Al, etc.

Rack Your Brain

Why do the conduction band and valence band overlap in metals?

2) Insulators:

- The energy band diagram of an insulator is shown below in figure 1.10.

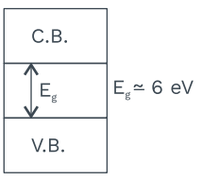


Fig. 1.10 Energy Band Diagram of Insulators

- Insulators have a very large forbidden energy gap $(\simeq 6\,eV)$.
- The electron in the valence band cannot gain enough energy to cross the forbidden energy gap and reach the conduction band.
- Since an insulator does not have any free electron in its conduction band. Hence, they do not conduct at room temperature.
- However, an insulator may conduct if its temperature is very high or a very high voltage is applied across it. This is known as the breakdown of the insulator.

For example, Wood, Diamond, etc.

3) Semiconductors:

- Figure 1.11 shows the energy band diagram of a semiconductor.

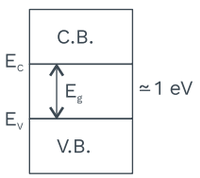


Fig. 1.11 Energy Band Diagram of a Semiconductor

- At '0'K, the semiconductor material acts as an insulator and has the same energy band diagram. The only difference is in the size of the band gap E_g, which is relatively smaller in a semiconductor $(\simeq 1eV)$ than the insulator $(\simeq 6\,eV)$.

Previous Years' Question

Which one of the following statements is correct?

a) For insulators, the band gap is narrow as compared to semiconductors.

b) For insultators the band gap is relatively wide, whereas for semiconductors it is narrow

c) The band gap is narrow in width for both the insulators and conductors

d) The band gap is equally wide for both conductors and semiconductors.

Rack Your Brain

Why is silicon preferred over germanium?

- The energy gap in a semiconductor is small, and hence, when the electrons are supplied with the thermal or optical energy, it jumps the energy gap and moves from the valence band to the conduction band.
- The conductivity of semiconductors can be tuned according to our needs.

Example: Si and Ge.

EFFECT OF TEMPERATURE ON ENERGY BAND GAP

- The energy gap decreases with an increase in temperature by the following relation:

$$E_{g(T)} = E_{g0} - \beta T \quad \ldots(1.12)$$

Where,

$E_{g(T)}$ = Energy gap at temperature T(in K)

E_{g0} = Energy gap at '0' Kelvin

T = Temperature (in Kelvin)

β = Material constant (in eV/K)

For Si, $\beta = 3.60 \times 10^{-4}$ eV/K

For Ge, $\beta = 2.23 \times 10^{-4}$ eV/K

Rack Your Brain

How can you justify that the energy gap decreases with an increase in temperature theoretically? (Explain without the help of any mathematical equation)

Example 1:

Find the energy gap of silicon at T = 300 K.

a) 1.4 eV **b)** 0.72 eV

c) 1.2 eV **d)** 1.1 eV

Sol:

At room temperature T = 300 K

For Si, $\beta = 3.60 \times 10^{-4}$ eV/K

$E_{g0} = 1.21$ eV

$E_{g(T)} = E_{g0} - \beta T$

$E_{g(T)} = 1.21 - 3.60 \times 10^{-4} \times 300$

$E_{g(T)} = 1.102$ eV

Example 2:

Find the energy gap of Ge at 400 K.

a) 0.71 eV **b)** 0.69 eV

c) 0.61 eV **d)** 0.64 eV

Sol:

$T = 400\ K$

$E_{g0} = 0.785\ eV$

For Ge, $\beta = 2.23 \times 10^{-4}\ eV/K$

$E_{g(T)} = E_{g0} - \beta T$

$E_{g(T)} = 0.785 - 2.23 \times 10^{-4} \times 400$

$E_{g(T)} = 0.695\ eV$

ENERGY vs MOMENTUM DIAGRAM

Wave vector $= \frac{2\pi}{\lambda}$

Let $\frac{2\pi}{\lambda} = k$

$$\hbar = \frac{h}{2\pi}$$

$$\hbar k = \frac{h}{2\pi} \cdot \frac{2\pi}{\lambda}$$

$\hbar k = p$

where, p = momentum & h = Planck's constant

$$k = \frac{p}{\hbar}$$

$k \propto$ momentum of an electron as $\hbar$ is a constant.

Kinetic energy $= \frac{p^2}{2m}$

$$E = \frac{\hbar^2 k^2}{2m} \quad ...(1.13)$$

$E \propto k^2$

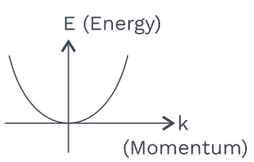
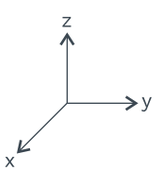


Fig. 1.12 Represents E vs k Diagram for a Free Electron to Move in Vacuum or Free Space.

- But any real crystal has 3-D periodic arrangements of atoms.



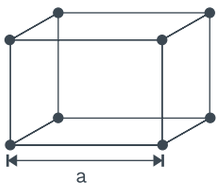
Previous Years' Question

Which one of the following is the correct relationship between the band gap of a material used in a photodetector and the energy of the incident photon?

a) $E_g \geq hc / \lambda$ **b)** $hv^2 / \lambda \geq E_g$

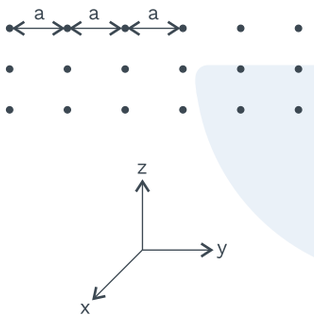
c) $hv \geq E_g$ **d)** $1/2hv \leq E_g$

(The symbols have usual meanings)



a = Lattice constant (Physical dimension of a unit cell in a crystal lattice).

- Arrangement of the atoms in a semiconductor is shown below:



- Atoms are present in the 3-dimensions.
- To make the things simpler, let us assume a single direction.

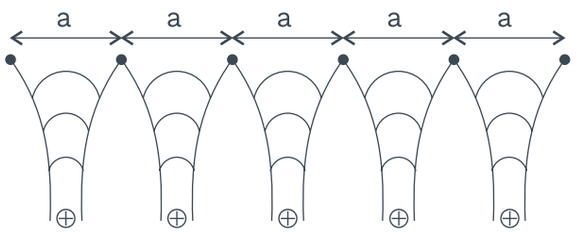


Fig. 1.13 Periodic Potential Well

- This periodic potential well has been set up by the atoms, and the electrons have to move through this periodic potential well.
- The electrons get scattered by the atoms and present at regular periodic intervals in all three directions. Hence, the E vs k diagram gets distorted.

Definitions

Crystalline Material:
"A crystalline material consists of a primarily organised crystal structure. A crystal is a solid composed of atoms, ions, or molecules arranged in a pattern that is repetitive in three dimensions. Each crystal structure within a specific crystal system is defined by a unit cell."

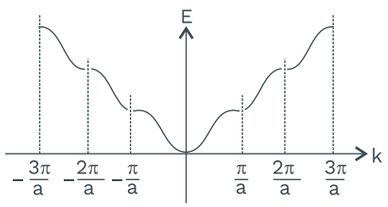


Fig. 1.14 Distorted E vs k Diagram.

- Consider only the first zone, i.e. first Brillouin zone.

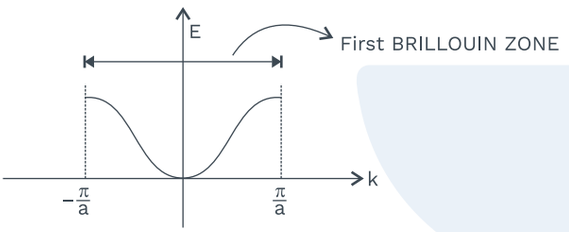


Fig. 1.15 E vs k Diagram of an Electron that Moves Inside a Semiconductor.

Note:

This is not an exact parabola, but near k = 0 it is exactly a parabola.

$E = \frac{\hbar^2 k^2}{2m}$ holds true for distorted parabola only when $m = m^*$

where m^* = effective mass of an electron.

EFFECTIVE MASS

- It is the mass which appears to have while responding to forces or interacting with other identical particles in a thermal distribution.
- Since a semiconductor has so many imperfections and periodic potential, hence the mass of the electron (e^-), which moves inside the semiconductor, is different from the rest mass (mass in a vacuum) of the electron.
  Free electron mass in the vacuum, $m_0 = 9.1 \times 10^{-31}$ kg.
  Effective mass, $m^* \neq m_0$

Now, $E = \frac{\hbar^2 k^2}{2m^*}$

$$\frac{dE}{dk} = \frac{\hbar^2 k}{m^*}$$

$$\frac{d^2E}{dk^2} = \frac{\hbar^2}{m^*}$$

$$m^* = \frac{\hbar^2}{\left(\frac{d^2E}{dk^2}\right)} \quad \text{...(1.14)}$$

- Expression $\frac{d^2E}{dk^2}$ represents the curvature of the graph.
- Entire band structure information is in $\frac{d^2E}{dk^2}$ expression. Now we don't need to solve quantum mechanics equations.
- All the classical physics equations can be applied to the electron (e^-) as long as we use effective mass (m^*).

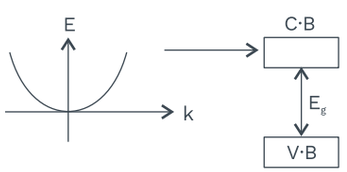


Fig. 1.16 E vs k Diagram for an e^- Corresponding to the Conduction Band.

Similarly,

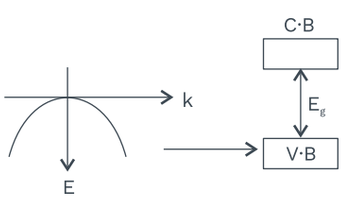


Fig. 1.17 E vs k Diagram for a Hole Corresponding to the Valence Band.

Example 3:
In the following figures, which figure will have the higher effective mass of an electron?

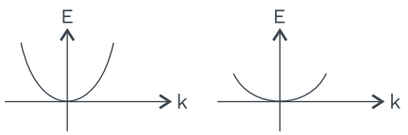


Figure A) Figure B)

a) Both have an equal effective mass
b) Figure a) will have a higher effective mass
c) Figure b) will have a higher effective mass
d) Insufficient data

Sol: We know that:

$$m^* = \frac{\hbar^2}{\left(\frac{d^2E}{dk^2}\right)}$$

$$\text{Effective mass } m^* \propto \frac{1}{\text{Curvature of the graph}}$$

Figure b) has lower curvature. Hence, figure b) will have a higher effective mass.
Hence, the correct option is c).

Previous Years' Question

The unit of $\frac{q}{KT}$ are:

a) V **b)** V^{-1}
c) J C **d)** J / K

DIRECT BAND GAP AND INDIRECT BAND GAP SEMICONDUCTORS

1) Direct band gap semiconductors:

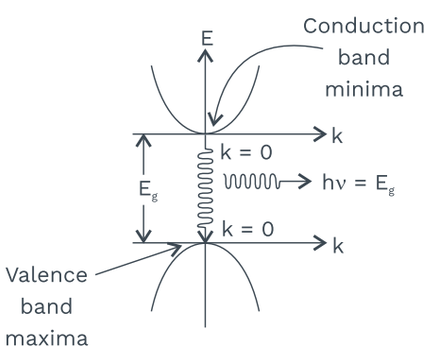


Fig. 1.18 E vs k Diagram of Direct Band Gap Semiconductor

- The valence band maxima and the conduction band minima of a direct band gap semiconductor occur at the exact value of k, i.e. k = 0.
- In a direct band gap semiconductors, an electron can make the minimum energy transition from the conduction band to the valence band without changing the k value.

  An electron in the conduction band can fall to an empty state in the valence band in a direct band gap semiconductor, releasing the energy difference (E_g) as a photon.

  Example: GaAs, GaN, etc.

2) Indirect band gap semiconductor:

- Indirect band gap semiconductor has its valence band maxima at a different value of k than its conduction band minima.

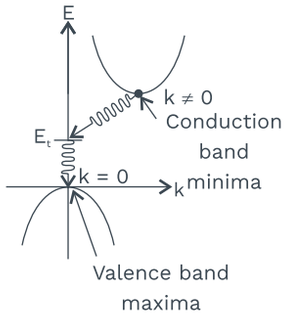


Fig. 1.19 E vs k Diagram of Indirect Band Gap Semiconductor

- The movement of an electron from the conduction band to the valence band needs a change in k value, i.e., the transition of an electron from the conduction band to the valence band requires a change in momentum.
- In an indirect band gap semiconductor, an electron in the conduction band cannot fall directly to the valence band. The electron falls to an intermediate state (E_t), where the electron experiences a change in momentum and energy.

Grey Matter Alert!

Optical devices such as LEDs and semiconductor lasers are made with direct band gap semiconductors, whereas silicon and germanium, which are indirect band gap semiconductor, is not used for optical devices.

- In an indirect transition, part of the energy is released as heat to the lattice.

For example: Si, Ge, etc.

INTRINSIC SEMICONDUCTORS

- A semiconductor crystal with no impurities or lattice defects is called an intrinsic semiconductor.
- There is no charge carrier at '0'k which means there are electrons present in the valence band while the conduction band is empty.
- Si and Ge are examples of intrinsic semiconductors.
- Both the elements belong to the IVth group of the periodic table and their atomic numbers are 14 and 32, respectively. Their valency is 4, i.e. both the elements have four electrons present in their outer shell.

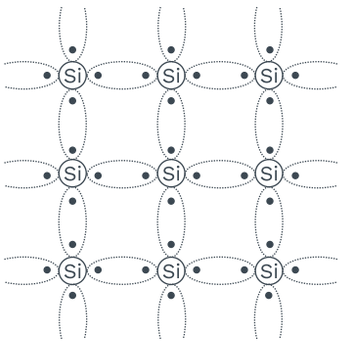


Fig. 1.20 Crystal Structure of Si, Illustrated Symbolically in Two Dimensions.

- Each atom in a silicon crystal contributes four valence electrons, so that the atom is tetravalent.
- Each silicon atom shares its valence electron with its four neighbouring atoms.
- There is the availability of four valence electrons, but the conductivity is low as the valence electrons are bound to the nucleus.
- At a very low temperature (0 Kelvin), no free carriers of electricity are available. Hence, the crystal behaves as an insulator.

Rack Your Brain

Why is carbon not used as a semiconductor, even though carbon is present in the IVth group of the periodic table?

- At room temperature, some of the covalent bonds will break because of the thermal energy supplied to the crystal (or if we supply a reasonable amount of thermal/optical energy, the bonds will break) and conduction is made possible. The minimum energy required to break the covalent bond is equal to the band gap energy (E_g).

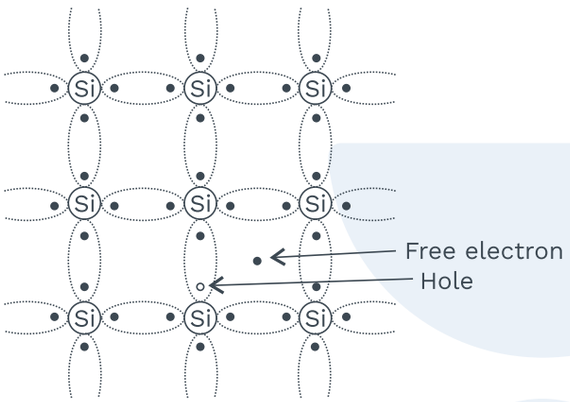


Fig. 1.21 Silicon Crystal Structure With a Broken Covalent Bond

Note:

- When a covalent bond breaks, an electron-hole pair is created.
- The concentration of electrons in the conduction band is equal to the concentration of holes in the valence band. Each carrier concentration is known as intrinsic semiconductor concentration (n_i).
- Thus, for an intrinsic semiconductor:
  $n = p = n_i$

CONCEPT OF HOLE

- When a covalent bond breaks, there will be an absence of an electron in that covalent bond, and that vacancy is called a hole.
- The hole may also act as a carrier of electricity.

Previous Years' Question

The minimum energy of a photon required for intrinsic is equal to

a) The energy of the bottom of the conduction band
b) The energy of the top of valence band
c) Forbidden gap energy
d) Fermi energy

- Suppose a photon of energy hν (which is $\geq E_g$) is incident on the crystal and breaks a covalent bond. Subsequently, an electron will jump to the conduction band, called a free electron, and a hole will be created in the valence band.

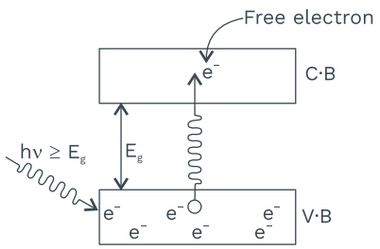


Fig. 1.22 Electron-hole Pair Generation (EHP Generation)

- The conductivity of a hole is as follows:
  1) When a hole is present in the covalent bond, it is easy for a valence electron in the neighbouring atom to leave its covalent bond to fill the hole.
  2) The movement of an electron from one covalent bond to another is shown in figure 1.23 b).

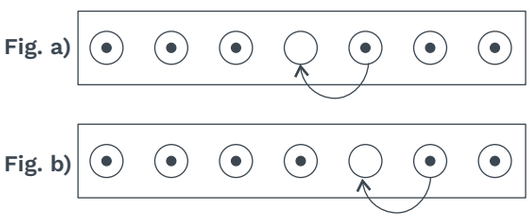


Fig. 1.23 Hole Movement in the Valence Band

  3) Electrons are moving in the left direction, but it seems like a hole is moving in the right direction. As a result, the hole moves in the opposite direction as the electron.
  4) As a result, we have a mechanism for current flow that involves a hole but does not involve a free electron.
  5) In an electric current, the hole acts as a positive charge, and its magnitude is the same as that of the charge of an electron.

6) The movement of holes results in the flow of current.

Example 4:
Which of the following statement (s) is/are correct?
A 'hole' in a semiconductor has:

a) Positive charge is equal to an electronic charge.
b) Positive mass equal to the mass of an electron.
c) An effective mass greater than the effective mass of an electron.
d) Negative mass and positive charge equal to the charge in the nucleus.

Sol:

- Hole has a positive charge equal to the charge of an electron.
- Effective mass of the hole is greater than the effective mass of an electron.

Hence, the correct options are a) and c).

Rack Your Brain

"Holes are less mobile electrons." Justify the statement.

CONDUCTION IN SEMICONDUCTOR

- When the electron-hole pair is generated, it creates two charged particles. One of them is electron having mobility (μ_n), and the other is the hole with mobility (μ_p). The current due to the movement of two charge carriers is in the same direction.
- Therefore, the conduction in semiconductors can be described by:

$$\sigma = (n\mu_n + p\mu_p)e \quad \ldots(1.15)$$

and for intrinsic semiconductor,

$$n_i = n = p \quad \ldots(1.16)$$

Accordingly,

$$\sigma_i = n_i(\mu_n + \mu_p)e \quad \ldots(1.17)$$

Equation 1.17) gives conductivity of the intrinsic semiconductor.

Where

n = Magnitude of concentration of electrons

p = Magnitude of concentration of holes

σ = Conductivity

n_i = Intrinsic concentration

e = Electronic charge = 1.6×10^{-19} C

- Intrinsic concentration of the pure type semiconductor differs under the following relationship.

$$n_i^2 = A_0 T^3 \exp\left(\frac{-E_{g_0}}{kT}\right) \quad ...(1.18)$$

Where,

Eg_0 = Energy gap at '0' K (in eV)

T = Absolute temperature in K

A_0 = Constant of material independent of temperature.

k = Boltzmann's constant in eV/K

- Hence, while increasing the temperature, n_i increases which ultimately increases the conductivity.
- Also, conductivity is the reciprocal of a material's resistivity.

$$\sigma = \frac{1}{\rho} \text{ and } \rho = \frac{RA}{l} \quad ...(1.19)$$

Where

σ = Conductivity of a material

ρ = Resistivity of a material

l = Length of material

A = Cross-sectional area of material

Definitions

Conductivity: "It is a measure of how easily an electric charge can pass through a material." **Conductor:** "It is a material that provides very little resistance to the flow of an electric current."

Example 5:

Evaluate the value of the intrinsic concentration of the silicon sample, which is kept at 500 K. (Assume $N_C = 2.6 \times 10^{19}$ /cm^3; $N_V = 1.24 \times 10^{19}$ /cm^3 and E_g = 1.2 eV at 300 K).

Sol:

According to the temperature and intrinsic concentration relationship formula,

$$n_i^2 = A_0 T^3 \exp\left[\frac{-E_{g_0}}{kT}\right]$$

Here, T = 500 K

$$n_i^2 = N_C N_V \exp\left(\frac{-E_g}{kT}\right)$$

$$= (2.6 \times 10^{19})(1.24 \times 10^{19})\left(\frac{500}{300}\right)^3 \exp\left[\frac{-1.2}{0.0259}\left(\frac{300}{500}\right)\right]$$

$$\Rightarrow 3.224 \times 10^{38}\left(\frac{125}{27}\right)\exp\left[\frac{-1.2}{0.0259}\left(\frac{3}{5}\right)\right]$$

$$\Rightarrow (14.9259 \times 10^{38})\exp\left[-\frac{36000}{1295}\right]$$

$\Rightarrow$ $(14.9259 \times 10^{38})\ \exp[-27.799]$

$\Rightarrow$ $n_i^2 = 1.2617933 \times 10^{27}$

and $n_i = \sqrt{1.2617 \times 10^{27}}$

$n_i = 3.552 \times 10^{13}\ cm^{-3}$

n_i is the required value of intrinsic semiconductor concentration.

Example 6:

The variation in intrinsic carrier concentration of a semiconductor:

1) Increases non-linearly with an increase in temperature.

2) Increases linearly with the increase of temperature value.

3) Increases exponentially with the decrease in band gap of the semiconductor.

Which of the above statements are correct?

a) 1), 2) and 3) **b)** 1) and 3)

c) 1) and 2) **d)** 1) only

Sol:

The correct option is b).

Previous Years' Question

An n-type semiconductor is illuminated by a steady flux of photons with energy greater than the band gap energy. The change in conductivity is:

a) $\Delta\sigma = 0$

b) $\Delta\sigma = e\left(\mu_n + \mu_p\right)\Delta n$

c) $\Delta\sigma = e\left(\mu_n \Delta n - \mu_p \Delta p\right)$

d) $\Delta\sigma = e\mu_n \Delta n$

DOPED / EXTRINSIC SEMICONDUCTOR

- Doped semiconductors, also known as impure or extrinsic semiconductors, contain a small amount of pentavalent or trivalent impurity atoms.

Extrinsic semiconductor:

- At room temperature, generally, conductivity of pure/intrinsic semiconductor is low. So, to increase the electrical conductivity of an intrinsic semiconductor, doping is done, which is the addition of impurity in the process of crystallization.
- The addition of impurity is very less of the order of one atom per million atoms of a pure intrinsic semiconductor.
- Hence, such a semiconductor with impurity atoms is defined as an impure or extrinsic semiconductor, and the process of adding impurity atoms is known as doping.

Doping:

- The doping material can be either pentavalent atoms (5 valence electrons) or trivalent atoms (3 valence electrons).

Examples of pentavalent atoms: Bismuth, Antimony, Arsenic, Phosphorus, etc.

Examples of trivalent atoms: Gallium, Indium, Aluminium, Boron, etc.

TYPE OF DOPING:

1) Donor (n-type) impurity:

- In this case, a dopant has five valence electrons, out of which 4 make covalent bonds, and 1 electron is left over, which is donated to the conduction band of the pure semiconductor.

Since it donates an electron, it is called as a donor atom.

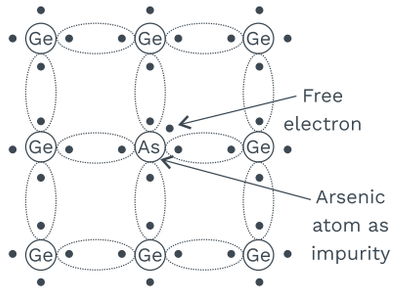


Fig. 1.24 Crystal Lattice with One Germanium Atom Displaced by Arsenic Atom

- The free electron is loosely bonded to the nucleus, and hence, only a small part of the energy will be enough to move it into the conduction band. (For Si – the energy required is 0.05 eV, and for Germanium Ge – Energy required is 0.01 eV).
- "n-type semiconductor" is another name for this type of doped atom, where a majority of carriers are electrons, and holes are minority charge carriers.

$$\therefore n = n_0 = N_C \exp\left(\frac{-(E_C - E_F)}{kT}\right) \quad ...(1.20)$$

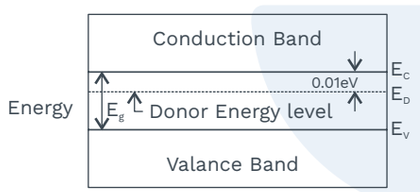
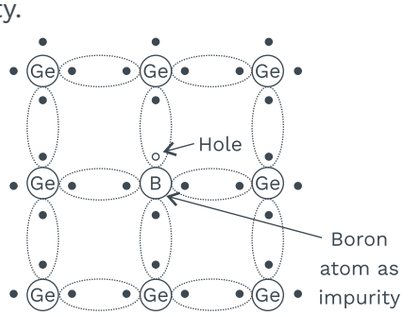


Fig. 1.25 Energy Band Diagram of n-type Semiconductors

- Here, for Ge, 0.01 eV energy is required for the electron to move to the conduction band.

2) Acceptor (p-type) impurity:

- In this type of impurity, the dopant has 3 valence electrons (e.g. B, Al, Ga and In), out of which all 3 electrons are used to make covalent bonds, but 1 covalent bond is left out, which creates a vacancy to accept an electron. This generates a hole which creates a possibility to accept a free electron, and hence, it is called an acceptor impurity.



Previous Years' Question

The resistivity of a uniformly doped n-type silicon sample is $0.5\,\Omega - cm$. If the electron mobility (μ_n) is $1250 cm^2 / V - sec$ and the charge of an electron is 1.6×10^{-19} Coulomb, the donor impurity concentration (N_D) in the sample is:

a) $2 \times 10^{16} / cm^3$

b) $1 \times 10^{16} / cm^3$

c) $2.5 \times 10^{15} / cm^3$

d) $2 \times 10^{15} / cm^3$

Fig. 1.26 Crystal Lattice Structure of p-Type Semiconductor

- In the p-type semiconductor, the holes are the majority charge carriers, and electrons are the minority charge carriers.

$$\therefore P = P_0 = N_V \exp\left(\frac{-(E_F - E_V)}{kT}\right) \quad ...(1.21)$$

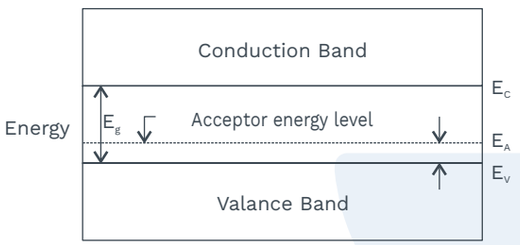


Fig. 1.27 Energy Band Diagram of p-type Semiconductor

ADVANTAGES OF DOPED SEMICONDUCTOR

1) Conductivity can be chosen by the amount of doping fraction.

2) Majority carriers (electrons or holes) can be chosen according to the application.

3) P-N junction diodes and transistor devices can be designed.

4) Doping fraction can vary and/or majority carrier within a piece of semiconductor easily.

FERMI LEVEL

Fermi energy is the highest energy value of electron at '0' K.

- Above the Fermi level, all the energy levels should be empty, and below the Fermi level, all the energy levels should be completely filled. (This is valid only for '0' K).

1) Fermi level in intrinsic semiconductor:

- In the case of pure semiconductor or intrinsic semiconductor, thermally generated electrons and hole concentration is maintained as constant. So, the fermi

Previous Years' Question

Due to illumination by light, the electron and hole concentrations in a heavily doped N-type semiconductor increase by Δn and Δp, respectively; if n_i is the intrinsic concentration then,

a) $\Delta n < \Delta p$ **b)** $\Delta n > \Delta p$

c) $\Delta n = \Delta p$ **d)** $\Delta n \times \Delta p = n_i^2$

level indicates at the midpoint of the forbidden energy gap, as shown in figure 1.28.

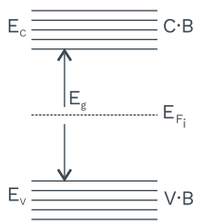


Fig. 1.28 Fermi Level in Intrinsic Semiconductor

2) Fermi level in n-type semiconductor:

- In an n-type semiconductor, donor level electron concentration will be in addition to the thermal generation. So the concentration of the electrons will be more than that of holes.
- So, the Fermi level should be indicated above the midpoint of forbidden energy gap as shown in the figure 1.29.

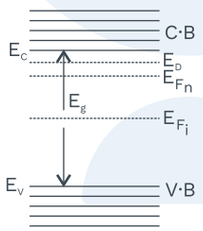


Fig. 1.29 n-Type Semiconductor's Fermi Level

3) Fermi level in p-type semiconductor

- In a p-type semiconductor, the acceptor hole concentration is present in addition to the thermally generated.
- Hole concentration will be more than that of electron concentration.
- So, the Fermi level should be indicated below the midpoint of the forbidden energy gap, as shown in figure 1.30.

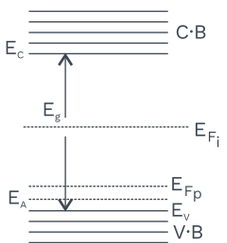


Fig. 1.30 Fermi Level in p-Type Semiconductor

- Exact location of the Fermi level depends on the doping level for a given temperature.

DEGENERATE TYPE SEMICONDUCTORS

1) Degenerate n-type semiconductor:

If the doping level of donor impurities is very high, then the Fermi level enters into the conduction band, which is known as a de-generate n-type semiconductor.

- Figure 1.31 shows the position of the Fermi level in de-generated n-type semiconductor.

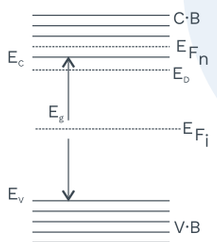


Fig. 1.31 Fermi Level Representation in the Degenerated n-Type Semiconductor

2) Degenerate p-type Semiconductor:

- If the doping level of acceptor impurities is very high, then the Fermi level enters into the valence band, which is known as the degenerated p-type semiconductor.
- Figure 1.32 shows the position of Fermi level in a de-generated p-type semiconductor.

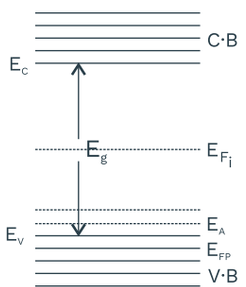


Fig. 1.32 Fermi Level Representation in Degenerate p-Type Semiconductor

Previous Years' Question

Consider the following statements for an n-type semiconductor:

1) E_F lies below E_D at a room temperature (T).

2) E_F lies above E_D as $T \rightarrow 0$

3) $E_F = E_D$ at some intermediate temperature

4) E_F is invariant with temperature

Where E_F is Fermi energy, and E_D is donor level energy. Which of these statement(s) is/are correct?

1) 1 and 2 **2)** 2 and 3
3) 4 only **4)** 1, 2 and 3

EFFECT OF TEMPERATURE ON THE FERMI LEVEL

1) Pure/Intrinsic semiconductor:

- When the temperature is increased, the thermally generated carrier (electrons and holes) also increases, but the concentration of electrons and holes will be the same. So the Fermi level must be at the midpoint of the forbidden energy gap only.
  Note: When the temperature increases, the position of the Fermi level remains same in the case of pure/intrinsic semiconductors.

2) n-type semiconductor:

- When the temperature goes on increasing gradually, the thermally generated carrier dominates the donor level electron and finally compared to the thermally generated carrier concentration, donor level electron concentration will be negligible.
- So, the fermi level gradually moves down and finally reaches the midpoint of the forbidden energy gap as in the case of an n-type semiconductor.

3) p-type semiconductor:

- When the temperature goes on increasing gradually, thermally generated carriers (electrons or holes) dominate the acceptor level holes and finally compared to the thermally generated carrier concentration, the acceptor level holes concentration will be negligible.
- So, the fermi level gradually moves upward and finally reaches the midpoint of the forbidden energy gap as in the case of p-type semiconductor.

CARRIER CONCENTRATIONS

- For the calculation of the semiconductor

Previous Years' Question

In a degenerate semiconductor, the majority of carriers are controlled by:
a) Fermi-Dirac statistics.
b) Maxwell-Boltzmann statistics.
c) Bose-Einstein (B-E) statistics.
d) Pauli's exclusion principle.

electrical properties, the number of the charge carriers per cm^3 in the material is required.

- One majority carrier is obtained for each impurity atom; the concentration of the majority carrier is in heavily doped material.
- The concentration of the minority carrier, however, is not obvious, nor is the temperature dependence of the carrier concentrations.
- In order to obtain an equation for carrier concentration, the distribution of carriers over the available energy states must be calculated using Fermi-Dirac statistics/distribution.

FERMI-DIRAC DISTRIBUTION

- In solids, electrons follow the Fermi-Dirac statistics.
- As a result of these statistics, the distribution of electrons at thermal equilibrium over a range of allowed energy levels is:

$$f(E) = \frac{1}{1 + e^{(E-E_F)/kT}} \quad ...(1.22)$$

Where

k = Boltzmann's constant

k = 8.62 × 10^{-5} eV/K or 1.38 × 10^{-23} Joule/K

E = Given energy level

E_F = Fermi energy level

T = Temperature (in kelvin)

Note: The Fermi-Dirac distribution function f(E) gives the probability of finding an electron at a given energy level.

- If an energy E, which is equal to the Fermi energy E_F, then the probability of finding an electron at energy level E_F is:

$\because E = E_F$

$$\therefore f(E_F) = \frac{1}{1 + e^{(E_F - E_F)/kT}}$$

$$f(E_F) = \frac{1}{1 + e^{\circ}}$$

Previous Years' Question

As the Fermi energy of silver is 8.8×10^{-19} J, the velocity of the fastest electron in silver at 0°K (Given, Rest mass of electron = 9.1×10^{-31}kg) is:

a) 3.33×10^5 m / s

b) 1.39×10^6 m / s

c) 4.40×10^7 m / s

d) 3×10^8 m / s

$$f(E_F) = \frac{1}{2}$$

Thus, an energy state at the Fermi level has a probability of $\frac{1}{2}$ (i.e. 50%) being occupied by an electron.

VARIATION OF f(E) WITH TEMPERATURE

At T = '0' K

- **Case i)**

  $(E < E_F)$

  $$f(E) = \frac{1}{1 + e^{(E-E_F)/kT}} \qquad \left(\because E < E_F \;\; \therefore E - E_F = -ve\right)$$

  $$f(E) = \frac{1}{1 + e^{-\infty}}$$

  $f(E) = 1 \Rightarrow 100\%$

- **Case ii)**

  $(E = E_F)$

  $$f(E) = \frac{1}{1 + e^{(E-E_F)/kT}} \qquad \left(\because E = E_F \;\; \therefore E - E_F = 0\right)$$

  $$f(E) = \frac{1}{1 + e^{0}}$$

  $$f(E) = \frac{1}{2} \Rightarrow 50\%$$

- **Case iii)**

  $(E > E_F)$

  $$f(E) = \frac{1}{1 + e^{(E-E_F)/kT}} \qquad \left(\because E > E_F \;\; \therefore E - E_F = +ve\right)$$

  $$f(E) = \frac{1}{1 + e^{\infty}}$$

  $f(E) = 0 \Rightarrow 0\%$

- f(E) indicates that at '0' K, the distribution takes the simple rectangular form.
- The distribution implies that at '0' K, every available energy state up to E_F is filled with electrons, and all states above E_F are empty.

Rack Your Brain

For which type of particle, Fermi -Dirac distribution is applicable?

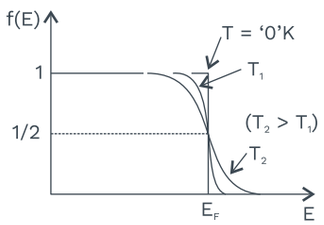


Fig. 1.33 Fermi-Dirac Distribution Function

- At a temperature higher than '0'K, some probability exists for states above the Fermi level to be filled.
- At T = T_1, there exist some probability f(E) which means that the states above E_F are filled, and there is a corresponding probability [1 – f(E)] which means that the states below E_F are empty.
- The Fermi function is symmetrical about E_F for all the temperatures, i.e., the probability $f(E_F+\Delta E)$, that a state ΔE above E_F is filled the same as the probability $[1 - f(E_F - \Delta E)]$, that a state ΔE below E_F is empty.
- The symmetry of the distribution of empty and filled states about E_F makes the Fermi level a natural reference point in the calculation of electron and hole concentration in the semiconductors.

ELECTRON AND HOLE CONCENTRATIONS AT EQUILIBRIUM

- If the densities of accessible states in the valence and conduction band are known, then the fermi distribution function can be used to compute the electron and hole concentrations in a semiconductor.
- The concentration of electrons in the conduction band is given by:

$$n_0 = \int_{E_C}^{\infty} f(E)\,N(E)\,dE \qquad ...(1.23)$$

where N(E)dE is the density of states (cm^{-3}) in the energy range dE. The n_0 (electron concentration) and p_0 (hole concentration) indicate the equilibrium conditions. The number of electrons per unit volume in the energy range dE is the product of the density of states and probability of occupancy f(E). Thus, the electron concentration is the integral over the entire conduction band as in equation 1.23).

$$\because N(E) = \frac{\sqrt{2}}{\pi^2}\left(\frac{m^*}{h^2}\right)^{3/2} E^{1/2}dE \qquad ...(1.24)$$

$$N(E) \propto E^{1/2}$$

Rack Your Brain

At T > 0K, the probability of a state with E > E_F filled is zero?
a) True **b)** False

- Since N(E) is proportional to $E^{1/2}$, the density of states in the conduction band increases with electron energy. On the other hand, the Fermi function becomes extremely small for large energies.
- The result is that the product f(E)N(E) decreases rapidly above E_C, and very few electrons occupy energy states above the conduction band edge.
- Similarly, the probability of finding an empty state (hole) in the valence band [1 – f(E)] decreases rapidly below E_V, and most holes occupy states near the top of the valence band.

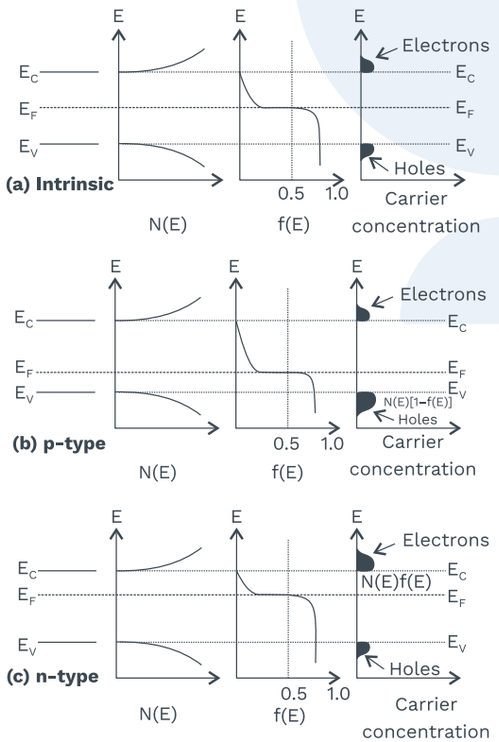


Fig. 1.34 Schematic Band Diagram, Density of States, Fermi-Dirac Distribution, and the Carrier Concentrations

Grey Matter Alert!

Applying a non-zero voltage to a device or illuminating it with light will cause a non-equilibrium condition, even if the temperature is constant.

- Therefore, in the conduction band, the electron concentration is simply the effective density of states at E_C times the probability of occupancy of E_C.

$$n_0 = N_C f(E_C) \quad ...(1.25)$$

- In this expression, we assume the fermi level E_F lies atleast several kT below the conduction band. The exponential term is large compared with unity, and the Fermi function $f(E_C)$ can be simplified as:

$$f(E_C) = \frac{1}{1+e^{(E_C-E_F)/kT}} \simeq e^{-(E_C-E_F)/kT} \quad ...(1.26)$$

- Since kT at room temperature is only 0.026eV, this is generally a good approximation. For this condition, the concentration of electrons in the conduction band is:

$$n_0 = N_C e^{-(E_C-E_F)/kT} \quad ...(1.27)$$

- As we know, the effective density of state N_C is:

$$N_C = 2\left(\frac{2\pi m_n^* kT}{h^2}\right)^{3/2} \quad ...(1.28)$$

Where,

h = Planck's constant

m_n^* = The density of state effective mass of electrons

- By similar arguments, the concentration of holes in the valence band is:

$$p_0 = N_V[1 - f(E_V)] \quad ...(1.29)$$

where N_V = Effective density of states in the valence band

- The probability of finding an empty state at E_V is:

$$1-f(E_V) = 1-\frac{1}{1+e^{(E_V-E_F)/kT}}$$

$$= \frac{1+e^{(E_V-E_F)/kT}-1}{1+e^{(E_V-E_F)/kT}}$$



Rack Your Brain

Does electron density increase the stability?

$$= \frac{1}{1+e^{-(E_V-E_F)/kT}} \simeq e^{-(E_F-E_V)/kT} \quad ...(1.30)$$

E_F is larger than E_V by several kT. From these equations, the concentration of holes in the valence band is:

$$p_0 = N_V e^{-(E_F-E_V)/kT} \quad ...(1.31)$$

- As we know, the effective density of states, N_V is

$$N_V = 2\left(\frac{2\pi m_p^* kT}{h^2}\right)^{3/2} \quad ...(1.32)$$

Where

h = Planck's constant

m_p^* = The density of states effective mass of holes.

- The electron and hole concentrations predicted by equations 1.27) and 1.31) are valid whether the material is intrinsic or doped, provided thermal equilibrium is maintained.
- Thus, for an intrinsic material, E_F lies at some intrinsic level E_i near the middle of the band gap [Figure 1.34-a)], and the intrinsic electron and hole concentrations are:

$$n_i = N_C e^{-(E_C-E_i)/kT} \quad ...(1.33)$$

and $$p_i = N_V e^{-(E_i-E_V)/kT} \quad ...(1.34)$$

- The product of n_0 and p_0 at equilibrium is a constant for a particular material and temperature, even if the doping is varied.

$$n_0 p_0 = \left(N_C e^{-(E_C-E_F)/kT}\right)\left(N_V e^{-(E_F-E_V)/kT}\right)$$

$$n_0 p_0 = N_C N_V e^{-(E_C-E_V)/kT}$$

$$n_0 p_0 = N_C N_V e^{-E_g/kT} \quad ...(1.35)$$

$$\left[\because E_C - E_V = E_g\right]$$

Grey Matter Alert!

In 1900, Max Planck empirically derived a formula for the observed spectrum. A photon's energy is equal to its frequency multiplied by the Planck's constant that constant is named in his honour.

Rack Your Brain

Which one of Si or Ge has a more effective density of states in the valence band at 300 K?

Similarly,

$$n_i p_i = N_C N_V e^{-E_g/kT} \quad ...(1.36)$$

- The intrinsic electron and hole concentrations are equal (since the carriers are created in pairs), $n_i = p_i$ thus, the intrinsic concentration is:

$$n_i = \sqrt{N_C N_V}\, e^{-E_g/2kT} \quad ...(1.37)$$

From equation 1.35), we get:

$$n_0 p_0 = n_i^2 \quad \{\text{where, } n_i^2 = N_C N_V e^{-E_g/kT}\} \quad ...(1.38)$$

- Comparing equations 1.33), 1.34) and 1.37), we can see that the intrinsic level E_i is the middle of the band gap ($E_C - E_i = E_g/2$), if the effective density of states N_C and N_V are equal.

Note: Effective density of states N_C and N_V are constant for a given semiconductor material at a fixed temperature. (Table 1.1 gives the values of effective density of states function and effective masses for Ge, Si and GaAs)

Type	$N_C(cm^{-3})$	$N_V(cm^{-3})$	m_p^*/m_0	m_n^*/m_0
Ge	1.04×10^{19}	6.0×10^{18}	0.37	0.55
Si	2.8×10^{19}	1.04×10^{19}	0.56	1.08
GaAs	4.74×10^{17}	7.0×10^{18}	0.48	0.067

Rack Your Brain

What is the dimensional formula of Planck's constant?

MASS ACTION LAW

- In a semiconductor material, under thermal equilibrium (at constant temperature), the product of electron and hole concentration is always a constant and equal to the square of intrinsic carrier concentration.

$$n \cdot p = n_i^2 \quad ...(1.39)$$

- The intrinsic carrier concentration n_i is a function of temperature and is given as:

$$n_i = \sqrt{A_0}\, T^{3/2} \exp[-E_g/2kT] \quad \ldots(1.40)$$

- This law is mainly used to calculate the concentration of minority charge carriers.
- In a p-type semiconductor, the holes are the majority charge carriers, and the electrons are the minority charge carriers.

For a p-type semiconductor:

$$n_p = \frac{n_i^2}{p_p} \quad \ldots(1.41)$$

- Similarly, in an n-type semiconductor, the electrons are the majority charge carriers, and holes are the minority charge carriers.

For an n-type semiconductor:

$$p_n = \frac{n_i^2}{n_n} \quad \ldots(1.42)$$

Where

p_n = hole concentration in n-type material.
n_n = electron concentration in n-type material.
n_p = electron concentration in p-type material.
p_p = hole concentration in p-type material.

- Conclusively, we can say that

$$\text{Minority charge carrier concentration} = \frac{n_i^2}{\text{Majority charge carrier concentration}}$$

But, the majority charge carrier concentration $\propto$ Doping concentration.

So,

$$\text{Minority charge carrier concentration} \propto \frac{1}{\text{Doping concentration}}$$

Hence,

Minority charge carrier concentration×Doping concentration = n_i^2 ...(1.43)

Note:

1) Mass action law does not apply to degenerative type semiconductors.
2) In a semiconductor, if the majority charge carrier concentration increases, the minority carrier concentration decreases. This is due to recombinations.

Example 7:

For a p-type semiconductor with $p = N_A$ and $n = \frac{n_i^2}{N_A}$, the electron concentration will fall below the intrinsic value because of some electrons.

a) Recombine with a hole

b) Virtually leave the crystal

c) Drop back to impurity states

d) Drop to donor impurity states

Sol:

The correct option is (a).

Example 8:

In an n-type silicon sample, the donor concentration at 300 K is $5 \times 10^{14}/cm^3$ corresponds to 1 impurity atom for 10^{18} silicon atoms. The electron and hole concentration in the sample will be: [Assume $n_i = 1.45 \times 10^{10}/cm^3$]

a) $n < 5 \times 10^{14}/cm^3$ and $p < 4.2 \times 10^5/cm^3$

b) $n = 5 \times 10^{14}/cm^3$ and $p = 4.2 \times 10^5/cm^3$

c) $n > 5 \times 10^{14}/cm^3$ and $p < 4.2 \times 10^5/cm^3$

d) $n < 5 \times 10^{14}/cm^3$ and $p > 4.2 \times 10^5/cm^3$

Sol:

- For an n-type semiconductor, the electron concentration $n > N_D$.

Let $n \simeq N_D$

Then, hole concentration

$$p = \frac{n_i^2}{N_D}$$

$$p = \frac{(1.45 \times 10^{10})^2}{5 \times 10^{14}}$$

$$p = 0.4205 \times 10^6/cm^3$$

$$p = 4.2 \times 10^5/cm^3$$

But, $n > N_D$

Hence, $n > 5 \times 10^{14}/cm^3$

Grey Matter Alert!

In 1864, Peter Waage and Cato Guldberg published a paper in which they propounded what has come to be known as the Law of Mass Action.

$$\therefore p < \frac{n_i^2}{N_D}$$

$$p < 4.2 \times 10^5/cm^3$$

Hence, the correct option is c).

Example 9:

Two identical pure semiconductors have been taken. One is doped with $10^{16}/cm^3$ numbers of the donor, and the other is doped with $10^{17}/cm^3$ numbers of acceptors. The minority carrier density in the first semiconductor specimen is $10^6/cm^3$. What is the minority carrier density in the other semiconductor specimen?

a) $10^6/cm^3$ **b)** $10^{17}/cm^3$ **c)** $10^{16}/cm^3$ **d)** $10^5/cm^3$

Sol:

According to mass action law:

$$n \cdot p = n_i^2$$

Hence, $n_1p_1 = n_2p_2$ (Taking $n \simeq N_D$ & $p \simeq N_A$)

$$10^{16} \times 10^6 = 10^{17} \times n_2$$

$$n_2 = \frac{10^{16} \times 10^6}{10^{17}}$$

$$n_2 = 10^5/cm^3$$

Hence, the correct option is d).

Rack Your Brain

How do you find the minority carrier concentration?

POSITION OF FERMI LEVEL IN PURE, n-TYPE, p-TYPE SEMICONDUCTORS

1) For intrinsic/pure semiconductor:

$$n = p = n_i$$

$$n = p$$

$$N_C e^{-(E_C - E_F)/kT} = N_V e^{-(E_F - E_V)/kT}$$

$$2\left[\frac{2\pi m_e kT}{h^2}\right]^{3/2} e^{(E_F - E_C)/kT}$$

$$= 2\left[\frac{2\pi m_h kT}{h^2}\right]^{3/2} e^{(E_V - E_F)/kT}$$

$$e^{(2E_F - E_C - E_V)/kT} = \left[\frac{m_h}{m_e}\right]^{3/2}$$

Taking natural log both sides, we get:

$$\frac{2E_F - E_C - E_V}{kT} = \ln\left(\frac{m_h}{m_e}\right)^{3/2}$$

$$2E_F - E_C - E_V = \frac{3}{2}kT \ln\left(\frac{m_h}{m_e}\right)$$

$$2E_F = E_C + E_V + \frac{3}{2}kT \ln\left(\frac{m_h}{m_e}\right)$$

$$E_F = \frac{E_C + E_V}{2} + \frac{3}{4}kT \ln\left(\frac{m_h}{m_e}\right) \quad \ldots(1.44)$$

Note: At T = 0, $E_F = \frac{E_C + E_V}{2}$; i.e. at the middle of the energy band gap.

2) For n-type semiconductor:

$$n = N_C e^{-(E_C - E_F)/kT}$$

In n-type semiconductor $n = N_D$

$$N_D = N_C e^{-(E_C - E_F)/kT}$$

$$\frac{N_C}{N_D} = e^{(E_C - E_F)/kT}$$

Taking natural log both sides, we get:

$$\ln\left(\frac{N_C}{N_D}\right) = \frac{E_C - E_F}{kT}$$

$$kT \ln\left(\frac{N_C}{N_D}\right) = E_C - E_F$$

$$E_F = E_C - kT \ln\left(\frac{N_C}{N_D}\right) \quad \ldots(1.45)$$

Note: From the above expression, we can say that

1) Fermi level is below the conduction band in n-type.

2) Fermi level depends on the doping concentration and the temperature.

Grey Matter Alert!

The doping concentration can be specified in different ways: the molar (atomic) percentage of the dopant, often specified in molar ppm (parts per million) the percentage by weight (or mass) of the dopant, often specified in ppm wt.

3) For p-type semiconductor:

$$p = N_V e^{-(E_F - E_V)/kT}$$

In p-type semiconductor $p = N_A$

$$N_A = N_V e^{-(E_F - E_V)/kT}$$

$$\frac{N_V}{N_A} = e^{(E_F - E_V)/kT}$$

Taking natural log both sides, we get:

$$\ln\left(\frac{N_V}{N_A}\right) = \frac{E_F - E_V}{kT}$$

$$E_F - E_V = kT \ln\left(\frac{N_V}{N_A}\right)$$

$$E_F = E_V + kT \ln\left(\frac{N_V}{N_A}\right) \quad \ldots(1.46)$$

Note: From the above expression, we can say the Fermi level lies above the valence band in the p-type semiconductor.

Rack Your Brain

Does Fermi level change with temperature?

Example 10:

What will be the position of the intrinsic Fermi level with respect to the center of the band gap in silicon at T = 300 K, if the effective carrier masses in silicon are $m_n^* = 1.06\ m_0$ and $m_p^* = 0.52\ m_0$.

Sol:

For intrinsic semiconductor

$$E_F = \frac{E_C + E_V}{2} + \frac{3}{4} kT \ln\left(\frac{m_h}{m_e}\right)$$

$$\because \quad \frac{E_C + E_V}{2} = E_{mid\,gap}$$

$$E_F - E_{mid\,gap} = \frac{3}{4} kT \ln\left(\frac{m_h}{m_e}\right)$$

$$= \frac{3}{4} \times 0.026 \ln\left(\frac{0.52 m_0}{1.06 m_0}\right)$$

$E_F - E_{mid\ gap} = -13.8$ meV

(Downward shift)

Rack Your Brain

How many Fermi levels are there in quasi-equilibrium?

Example 11:
Determine the values of n_0 and p_0 for silicon at temperature, T = 300 K if the Fermi energy is 0.22 eV above the valence band energy.

Given, $N_V = 1.02 \times 10^{19}/cm^3$

$N_C = 2.2 \times 10^{18}/cm^3$

Sol:

$\because \quad E_F - E_V = 0.22\,eV$

$$p_0 = N_V e^{-(E_F - E_V)/kT}$$

$$p_0 = 1.02 \times 10^{19} \times e^{(-0.22/0.026)}$$

$$p_0 = 2.15 \times 10^{15}/cm^3$$

$$n_0 = N_C e^{-(E_C - E_F)/kT}$$

$\because$ E_G = 1.1 eV for Si at T = 300 K.

and $E_F = E_V + 0.22$ (given)

$E_C - E_V = 1.1$

$E_C - (E_F - 0.22) = 1.1$

$E_C - E_F = 1.1 - 0.22$

$E_C - E_F = 0.88$ eV

Now, $n_0 = 2.2 \times 10^{18} \times e^{(-0.88/0.026)}$

$n_0 = 4.39 \times 10^3/cm^3$

CHARGE NEUTRALITY

- At thermal equilibrium, the semiconductor crystal is electrically neutral.
- According to the electrical neutrality condition, we can write:

Total positive charge = Total negative charge

$$\text{i.e., } N_D + p = N_A + n \qquad \ldots(1.47)$$

- We can convert an n-type semiconductor into a p-type semiconductor by counter doping it with acceptors such as $N_A > N_D$.

For n-type material:
According to charge neutrality condition,

$$N_D + p = N_A + n$$

For n-type material $N_A = 0$

$$p = \frac{n_i^2}{n_n} \quad (n_n = \text{density of electron})$$

$$\therefore N_D + \frac{n_i^2}{n_n} = n_n$$

$$n_n N_D + n_i^2 = n_n^2$$

$$n_n^2 - n_n N_D - n_i^2 = 0$$

$$n_n = \frac{N_D}{2} \pm \sqrt{\frac{N_D^2 + 4n_i^2}{4}} \qquad \left\{ n_n = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \right\}$$

[ignore –ve sign.]

$$\left[n_n = \frac{N_D}{2} + \sqrt{\left(\frac{N_D}{2}\right)^2 + n_i^2} \right] \qquad \text{...(1.48)}$$

When N_D is comparable with n_i.

- If $N_D >> n_i$

Then, $n_n = \frac{N_D}{2} + \sqrt{\left(\frac{N_D}{2}\right)^2}$

$n_n = N_D$

Similarly for p-type:

$$p_p = \frac{N_A}{2} + \sqrt{\left(\frac{N_A}{2}\right)^2 + n_i^2} \qquad \text{...(1.49)}$$

When N_A is comparable with n_i.

- If $N_A >> n_i$

Then, $p_p = \frac{N_A}{2} + \sqrt{\left(\frac{N_A}{2}\right)^2} \Rightarrow p_p = \frac{N_A}{2} + \frac{N_A}{2}$

$p_p = N_A$

Rack Your Brain

What is freeze-out in a semiconductor?

COMPENSATED SEMICONDUCTOR

- A compensated semiconductor material has both acceptor ions and donor ions in the same region.
- It can be performed by diffusing donor impurities into a p-type semiconductor or by diffusing acceptor impurities into an n-type semiconductor material.

- p-type semiconductor occurs if ($N_A > N_D$)

$$p_0 \simeq N_A - N_D$$

$$n_0 = \frac{n_i^2}{N_A - N_D}$$

- n-type semiconductor occurs if ($N_D > N_A$)

$$n_0 \simeq N_D - N_A$$

$$p_0 = \frac{n_i^2}{N_D - N_A}$$

Where

n_0 = Number density of electrons

p_0 = Number density of holes

1) Concentration of electron in a compensated n-type semiconductor:

By electrical neutrality conditions,

$$N_D + p = N_A + n$$

$$N_D + \frac{n_i^2}{n} = N_A + n$$

$$nN_D + n_i^2 = nN_A + n^2$$

$$-n^2 + n(N_D - N_A) + n_i^2 = 0$$

$$n^2 - n(N_D - N_A) - n_i^2 = 0$$

$$n = \frac{N_D - N_A}{2} + \sqrt{\left(\frac{N_D - N_A}{2}\right)^2 + \frac{4n_i^2}{4}}$$

$$n = \frac{N_D - N_A}{2} + \sqrt{\left(\frac{N_D - N_A}{2}\right)^2 + n_i^2} \quad \ldots(1.50)$$

Similarly, the concentration of holes in a compensated p-type semiconductor is

$$p = \frac{N_A - N_D}{2} + \sqrt{\left[\frac{N_A - N_D}{2}\right]^2 + n_i^2} \quad \ldots(1.51)$$

Example 12:
Consider a p-type silicon semiconductor at temperature, T = 300 K in which N_D=0 and $N_A = 10^{17}/cm^3$. Let n_i be the intrinsic carrier concentration, $n_i = 1.5\times10^{10}/cm^3$. Find the thermal equilibrium of electron and hole concentrations for the silicon sample.

Sol:
Given,

$n_i = 1.5 \times 10^{10}/cm^3$

$N_A = 10^{17}/cm^3$

$N_D = 0$

$\because$ It is a p-type semiconductor.

$$\therefore \quad p_0 = \frac{N_A - N_D}{2} + \sqrt{\left(\frac{N_A - N_D}{2}\right)^2 + n_i^2}$$

$$p_0 = \frac{N_A}{2} + \sqrt{\left(\frac{N_A}{2}\right)^2 + n_i^2}$$

$\because N_A >> n_i$

$\therefore N_A^2 >>> n_i^2$

$$\text{Hence, } p_0 = \frac{N_A}{2} + \sqrt{\left(\frac{N_A}{2}\right)^2}$$

$p_0 = N_A$

$p_0 = 10^{17}/cm^3$

By mass action law: $n_0 p_0 = n_i^2$

$$n_0 = \frac{n_i^2}{p_0}$$

$$n_0 = \frac{(1.5 \times 10^{10})^2}{10^{17}}$$

$n_0 = 2.25 \times 10^3/cm^3$

Rack Your Brain

Can electric field intensity be negative?

Example 13:
Consider a Ge semiconductor at T = 300 K in which $N_A = 3 \times 10^{15}/cm^3$ and $N_D = 10^{16}/cm^3$. Assume $n_i = 1.5 \times 10^{10}/cm^3$. Calculate the electron and hole concentrations at thermal equilibrium for the Ge semiconductor.

Sol:
Given,

$n_i = 1.5 \times 10^{10}/cm^3$

$N_A = 3 \times 10^{15}/cm^3$

$N_D = 10^{16}/cm^3$

$\because N_D > N_A$, hence it is an n-type compensated semiconductor.

$$n_0 = \frac{N_D - N_A}{2} + \sqrt{\left(\frac{N_D - N_A}{2}\right)^2 + n_i^2}$$

$$n_0 = \frac{10^{16} - 3 \times 10^{15}}{2} + \sqrt{\left(\frac{10^{16} - 3 \times 10^{15}}{2}\right)^2 + (1.5 \times 10^{10})^2}$$

$n_0 = 7 \times 10^{15}/cm^2$

From mass action law, $n_0 p_0 = n_i^2$

$$p_0 = \frac{n_i^2}{n_0}$$

$$p_0 = \frac{(1.5 \times 10^{10})^2}{7 \times 10^{15}}$$

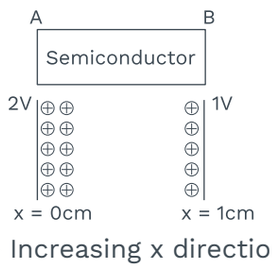
$p_0 = 3.214 \times 10^4/cm^3$

ELECTRIC FIELD INTENSITY

- Electric field intensity is defined as the negative rate of change of voltage for distance.

$$E = -\frac{dV}{dx}$$

- Let us assume a semiconductor whose one end (i.e. end A) is kept at 2V and x = 0 cm and another end (i.e. end B) is kept at 1 V and x = 1 cm.



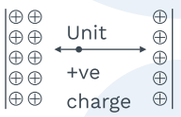
$$E = -\left(\frac{V_2 - V_1}{x_2 - x_1}\right)$$

$$E = -\left(\frac{1-2}{1-0}\right)$$

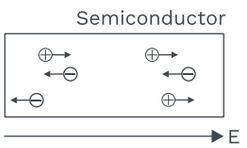
$E = 1$ V/cm

- Hence, E is +ve which means the direction of E is in the +x direction.

Note: The direction of E is the same as the direction of the force on a unit positive charge.



- Direction of an electric field is always from higher potential to lower potential.
- Negative sign signifies the direction of an electric field towards decreasing potential.
- On a positive charge, the direction of force is the same as the direction of an electric field.
- On a negative charge, the direction of force is the opposite to the direction of an electric field.
- Under the influence of electric field intensity E, positive and negative charge carrier will move in the opposite direction but contributes to the current in the same direction.



Direction of electric field = → rightwards
On +ve charges direction of force = → rightwards
Current due to +ve charge = → I_1 rightwards

On –ve charges direction of force = ← leftwards
Current due to –ve charge = → I_2 rightwards
Net Current = → I_1 + I_2 rightwards

DRIFT VELOCITY

The average velocity of charged particles in the material caused by an electric field is referred to as drift velocity. In general, electrons in a conductor will move at random, yielding an average velocity of zero. Drift occurs when an external electric field adds a small net flow in one direction to the random motion.
Drift velocity is proportional to an external electric field.

i.e. $V_d \propto E$

$V_d = \mu E$

Where μ is constant, known as mobility.

Note: For +ve charge particle $V_d = + \mu_p E$

For negative charge particle $V_d = - \mu_n E$

Where negative sign indicates that electrons move in the direction opposite to the electric field direction.

Rack Your Brain

What happens to the drift velocity when the length is doubled?

MOBILITY

Even at thermal equilibrium, charge carriers in the solids are in constant motion. At room temperature, an individual electron's thermal motion can be visualized as random scattering from lattice vibrations, impurities, other electrons, and defects. Because the distribution is random, there is no net movement of the group of n electrons/cm^3 over any given period.

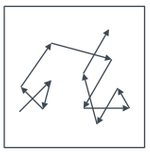


Fig. 1.35 Random Motion of an Electron in a Solid.

$\because v_d = \mu E$

$$\mu = \frac{v_d}{E} \rightarrow \frac{\text{cm/sec}}{\text{Volt/cm}} = \text{cm}^2/\text{V-sec}$$

Where

v_d = drift velocity

E = electric field intensity

μ = mobility

Unit of μ = cm^2/V-sec

If E = 1 V/cm

Then, $\mu = v_d$

- Hence, mobility is drift velocity under unit electric field intensity.

Mobility	Si	Ge	GaAs
μ_n	3800	1300	5800
μ_p	1800	500	400
$\left(\frac{\mu_n}{\mu_p}\right)$	2.1 : 1	2.6 : 1	14.5 : 1

Rack Your Brain

Which semiconductor has the highest electron mobility?

$v_d = \mu E$

$v_{d_1} = \mu_1 E$

$v_{d_2} = \mu_2 E$

If $\mu_1 > \mu_2$

Then, $v_{d_1} > v_{d_2}$

- High mobility value signifies a higher current contribution of the charge carrier.
  Note: If μ increases $\Rightarrow$ v_d increases $\Rightarrow$ Rate of flow of charge carrier increases $\Rightarrow$ Current increases.
- Electron mobility is always higher than the hole mobility in a particular semiconductor.

Example 14:
At one point, a small amount of minority carriers is injected into a homogeneous semiconductor crystal. A 10 V/cm electric field is applied across the crystal, which moves the minority carriers 1 cm in 20 μsec. What will be the mobility in (cm^2/v-sec)?

Sol:

We know that, Velocity $= \frac{\text{Distance}}{\text{Time}}$

$$v_d = \frac{1}{20 \times 10^{-6}} \text{ cm/sec}$$

$$v_d = 5 \times 10^4 \text{ cm/sec}$$

$$\because v_d = \mu E$$

$$\mu = \frac{v_d}{E}$$

$$\mu = \frac{5 \times 10^4}{10}$$

$$\mu = 500 \text{ cm}^2/\text{V-sec}$$

Rack Your Brain

Why is electron mobility higher than hole mobility?

EFFECT OF TEMPERATURE ON MOBILITY

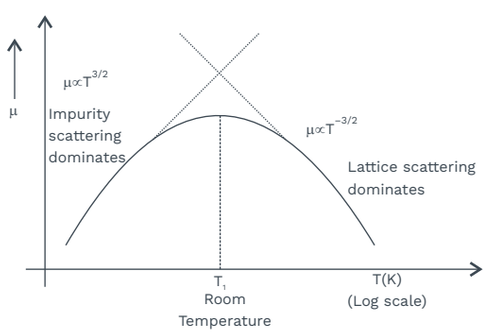


Fig. 1.36 Graph Between μ and T

- The effect of temperature on the mobility can be discussed in the following two ways:
  1) Impurity scattering
  2) Lattice scattering

1) Impurity scattering:
When temperature increases, the velocity of the electrons (e^{-}) also increases because of which scattering of electrons (e^{-}) by the impurity dose will be less so that the mobility increases with increase in temperature.

2) Lattice scattering:
In this case, when the temperature increases oscillation frequency for the atoms at the lattice point also increases. This causes e^{-} to experience more collisions per second which in turn causes mobility to decrease.
Note: When the temperature (by default room temperature) increase mobility decreases.

EFFECT OF IMPURITY ON MOBILITY

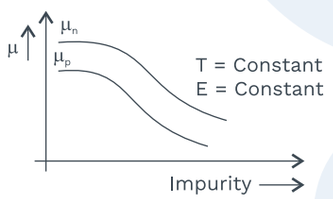


Fig. 1.37 Graph between mobility and impurity

- When the temperature and the electric field are constant, and the impurity is increased, then the mobility decreases.

VARIATION OF MOBILITY WITH ELECTRIC FIELD

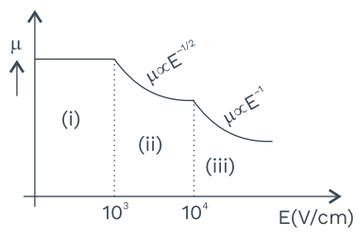


Fig. 1.38 Graph Between μ and E.

Rack Your Brain

What is the effect of temperature on the mobility and conductivity of holes and electrons?

Grey Matter Alert!

The mobilities caused by two or more scattering mechanisms are added inversely:

$$\frac{1}{\mu} = \frac{1}{\mu_1} + \frac{1}{\mu_2} + \frac{1}{\mu_3} + \ldots.$$

i) $0 < E < 10^3$ V/cm: (Low electric field region)

- In region 1), mobility is constant with respect to electric field intensity.
- At the low electric field region, μ is independent of electric field intensity.

ii) $10^3 < E < 10^4$ V/cm: (Moderate electric field region)

- In this region, mobility decreases with an increase in electric field, and that variation is non-linear.

$\mu \propto E^{-1/2}$

iii) $E > 10^4$ V/cm: (High electric field region)

- In this region, mobility decreases more rapidly with an increase in an electric field and that variation is also non-linear.

$\mu \propto E^{-1}$

VARIATION OF v_d WITH E

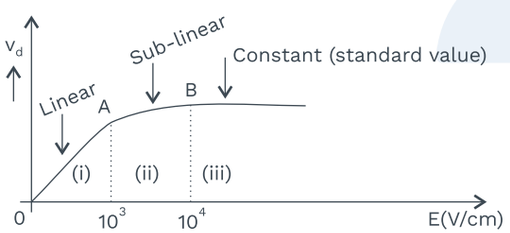


Fig. 1.39 Graph between v_d and E.

i) $0 < E < 10^3$ V/cm:

- When the electric field strength increases from the initial level, i.e. when $E < 10^3$ V/cm, drift velocity also increases in the same proportion, and this is shown in the graph from 0 to A.

$v_d = \mu E$

$\frac{dv_d}{dE} = \mu$, Linear variation

Rack Your Brain

When temperature increases, mobility decreases and conductivity increases. We also know that conductivity ∝ mobility. How will you justify the above statement?

ii) $10^3 < E < 10^4$ V/cm:

- When the electric field increases more than 10^3V/cm, the electron will get more kinetic energy, and the collision of an electron with the lattice point causes its energy to be lost in the form of a phonon. So the curve AB is sub-linear.

  $\because \mu \propto E^{-1/2}$

  $\mu = K_1 E^{-1/2}$

  $v_d = \mu E$

  $v_d = K_1 E^{-1/2} \cdot E$

  $v_d = K_1 E^{1/2}$
- Non-linear variation, but the rate of change is less.

iii) $E > 10^4$ V/cm:

- If we further increase the electric field, then the loss of phonon will be more in such a manner that drift velocity will be constant.

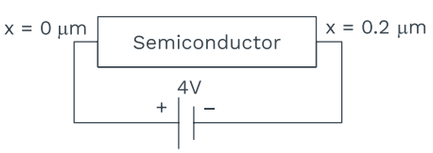
  $\because \mu \propto E^{-1}$

  $\mu = K_2 E^{-1}$
  $v_d = \mu E$
  $v_d = K_2 E^{-1} \cdot E$
  $v_d = K_2$
- The value of v_d gets saturated and becomes constant.
- At high electric field, drift velocity enters into saturation region and will not change with change in electric field intensity.
- In this region, v_d is independent of E and has a constant value and that value of v_d is known as saturation velocity.

Note: v_d can not attain a value greater than 10^7 cm/sec.

$$v_{d_{max}} = 10^7 \text{ cm/sec}$$

Example 15:



A semiconductor is shown in the figure with the applied potential of 4V. Mobility of e^- in the semiconductor is μ_n = 1350 cm^3/V-sec. Then the drift velocity of the electron in the semiconductor is:

a) 2.7×10^8 cm/sec

b) 10^7 cm/sec

c) 1.35×10^8 cm/sec

d) 2.7×10^6 cm/sec

Sol:

$$E = -\frac{dV}{dx}$$

$$E = -\frac{[0-4]}{[0.2\times10^{-4}-0]}$$

$$E = \frac{4\times10^4}{0.2}$$

$E = 2 \times 10^5$ V/cm

$v_d = \mu E$

$v_d = 1350 \times 2 \times 10^5$

$v_d = 2.7 \times 10^8$ cm/sec

Note: This value of v_d is not possible. Saturated value of v_d is the maximum value of v_d.

$$v_{d_{max}} = 10^7 \text{cm/sec}$$

Hence, the correct option is b).

CURRENT

The rate of flow of the charge carrier is called current.

1) Drift current:

('Drift' means by force)

The current that flows due to the electric field intensity is called as drift current.

2) Drift current density:

- Drift current per unit cross-sectional area is called as drift current density.
- Drift current density is given by:

  $J_{drift} = \sigma E$ A/cm^2

  σ = conductivity

E = Electric field intensity

$J_{drift} = J_{n_{drift}} + J_{p_{drift}}$

$J_{n_{drift}} = \sigma_n E$ (electron drift current density)

$J_{p_{drift}} = \sigma_p E$ (hole drift current density)

3) Diffusion and diffusion current:

- In a semiconductor, if at one end, the carrier concentration is high and on the other end, the carrier concentration is low, then the charge carriers have a tendency to flow from higher concentration to lower concentration. This process of flow is called as diffusion.
- The current that flows due to the diffusion process is called diffusion current.

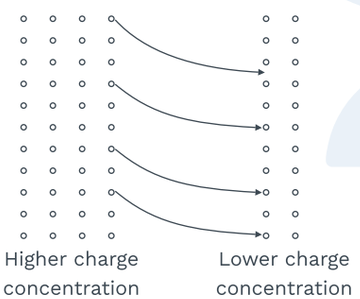


Fig. 1.40 Concentration Gradient

- For diffusion to occur, there must be a concentration of the gradient. Till the concentration gradient is there, diffusion will occur, and diffusion current will flow.
- The diffusion current crossing a unit area (the current density) is the particle flux density multiplied by the charge of the carrier:

$$J_{n(diff)} = -(-q)D_n \frac{dn(x)}{dx} = +qD_n \frac{dn(x)}{dx} \quad ...(1.52)$$

$$J_{p(diff)} = -(+q)D_p \frac{dp(x)}{dx} = -qD_p \frac{dp(x)}{dx} \quad ...(1.53)$$

- In a semiconductor, electrons and holes move

Grey Matter Alert!

Diffusion current obeys Fick's law and drift current follows Ohm's law.

together in a carrier concentration gradient, but the resulting currents are in opposite directions (as shown in the above equation) because of the opposite charge of electrons and holes.

$$J_{diff} = J_{n\,(diff)} + J_{p\,(diff)} \quad ...(1.54)$$

$$J_{(diff)} = qD_n \frac{dn(x)}{dx} - qD_p \frac{dp(x)}{dx} \quad ...(1.55)$$

- Total current density is the sum of drift current density and diffusion current density.

$$J_{total} = J_{drift} + J_{diff} \quad ....1.56)$$

$$J_n(x) = J_{n(drift)} + J_{n(diff)} \quad .....1.57)$$

$$J_n(x) = \underbrace{q\mu_n n(x)}_{drift} E(x) + \underbrace{qD_n \frac{dn(x)}{dx}}_{diffusion} \quad ...(1.58)$$

Similarly,

$$J_p(x) = q\mu_p p(x)E(x) - qD_p \frac{dp(x)}{dx} \quad ...(1.59)$$

- The total current density is the sum of the contributions due to electrons and holes.

$$J(x) = J_n(x) + J_p(x) \quad ...(1.60)$$

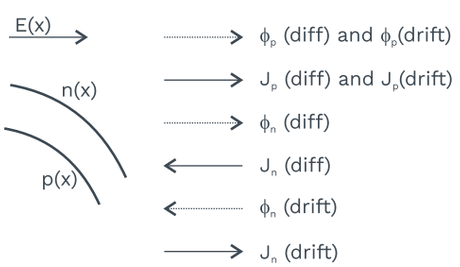


Fig. 1.41 Direction of Drift and Diffusion in a Carrier Gradient

4) Diffusion length:

When the charge carrier gets diffused, they also recombine with the opposite charge carrier. The

Previous Years' Question

The Hall constant in a p-Si bar is given by $5 \times 10^3 cm^3$ / Coulomb. The hole concentration in the bar is given by,

a) 1.00×10^{15} / cm^3

b) 1.25×10^{15} / cm^3

c) 1.50×10^{15} / cm^3

d) 1.6×10^{15} / cm^3

average distance travelled by the charge carrier before recombination is called the length of diffusion.

- Length of diffusion is given by:

$$L = \sqrt{D\tau} \quad \text{...(1.61)}$$

Where

τ = carrier life-time

D = Diffusion constant.

For electron, $L_n = \sqrt{D_n \tau_n}$

where

τ_n = electron life-time

D_n = Diffusion constant for electron

For hole, $L_p = \sqrt{D_p \tau_p}$

Where,

τ_p = hole life-time

D_p = Diffusion constant for hole

MINORITY CARRIER INJECTION/INJECTION BY ILLUMINATION

- When holes are injected at one end of an n-type semiconductor by the process of diffusion, this concentration will go on decreasing and is given by the relation of

$$P(x) = P_{n_0} + P'_{n_0} e^{-x/L_p} \quad \text{...(1.62)}$$

Where P_{n_0} = Thermally generated hole concentration

P'_{n_0} = Injected hole (carrier) concentration (concentration generated by optical excitation)

x = Length along the semiconductor

L_p = Diffusion length of hole

Grey Matter Alert!

The diffusion length ranges from a fraction of a micrometre to 100's of micrometres

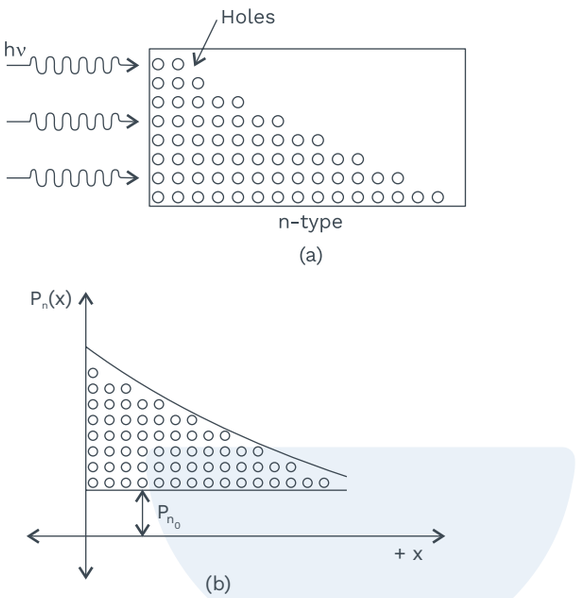


Fig. 1.42 Diffusion of Holes in n-Type Semiconductor

EINSTEIN RELATION TO SEMICONDUCTOR

- It states that the higher the value of the mobility of the charge carrier, the higher will be the tendency to diffuse.
- This gives the relation between mobility 'μ' and the diffusion constant D.

$$\frac{D_n}{\mu_n} = \frac{D_p}{\mu_p} = \frac{kT}{q} = V_T \quad ...(1.63)$$

HALL EFFECT

- The Hall effect is caused by the forces exerted on a moving charge by an electric and magnetic field.

Statement: "When a current-carrying specimen (metal or semiconductor) is placed in a transverse magnetic field (B), an electric field (E) is induced in the direction perpendicular to both current (I) and magnetic field (B)". This phenomenon is known as the Hall effect.

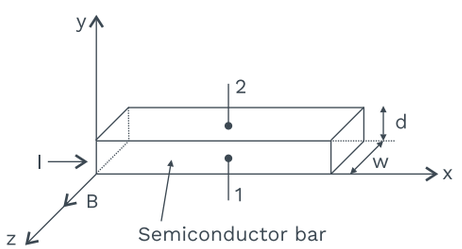


Fig. 1.43 A Hall Effect Experiment

- Figure 1.43 illustrates the Hall effect. The current (I) is in the positive x-direction, and the magnetic field b) is in the positive z-direction.
- Electrons and holes moving through the semiconductor will be subjected to a force. The force acts on the charged particle moving in the magnetic field.
- The force is in the negative y-direction independent of whether the carriers are holes or electrons.
- This charge accumulation induces an electric field in the y-direction.
- As a result, a potential, known as the Hall voltage, appears between surfaces 1 and 2.
- In a steady-state, the magnetic field force will be balanced by the induced field force.

$$qE = Bqv \quad \ldots(1.64)$$

Where q is the magnitude of the charge on the carrier and v is the drift speed.

- E is given by:

$$E = \frac{V_H}{d} \quad \ldots(1.65)$$

Where d is the distance between surfaces 1 and 2. From equations 1.64) and 1.65), we get:

$$\frac{V_H}{d} = Bv$$

Since, $J = \frac{I}{A} = \rho v$

$$v = \frac{I}{\rho A}$$

[where A is the cross-section area and is equal to w·d]

So, $\frac{V_H}{d} = \frac{BI}{\rho wd}$

$$V_H = \frac{BI}{\rho w} \quad \ldots(1.66)$$

where

w is the width of the specimen.

ρ is the charge density.

Previous Years' Question

For a particular material, the Hall the coefficient is found to be zero. The material is

a) Intrinsic semiconductor
b) Extrinsic semiconductor
c) Metal
d) Insulator

- Hall coefficient R_H is determined by:

$$R_H = \frac{1}{\rho} = \frac{V_H w}{BI} \quad ...(1.67)$$

- If the conductivity is measured together with the Hall coefficient, the mobility can be determined from:

$$\mu = \sigma R_H \quad ...(1.68)$$

If random thermal distribution is taken into account, then R_H and μ becomes:

$$R_H = \frac{3\pi}{8\rho} \quad ...(1.69)$$

$$\text{and} \quad \mu = \left(\frac{8\sigma}{3\pi}\right) \cdot R_H \quad ...(1.70)$$

Importance of Hall-effect:

1) Hall effect is used to determine the type of semiconductor.

2) It is used to measure the majority carrier concentration.

3) It is used to measure the majority of carrier mobility.

Example 16:

Consider the given figure of an n-type semiconductor bar:

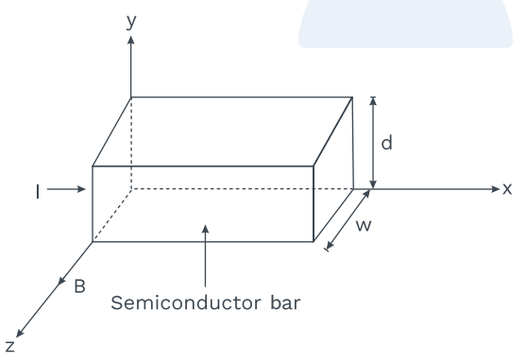


Fig. 1.44 n-Type Semiconductor Bar

The parameters for the Hall experiment setup are:

Electric field = 500 V/m

B_z = 10.5 Wb/m^2

w = 10 mm

d = 5 × 10^{-2} m

Mobility: μ_n= 1.2 m^2/V-sec

The Hall voltage (V_H) is _____ volts.

Sol:

Hall voltage:

$$V_H = \frac{I_x \cdot B_z}{\rho w}$$

As, $I_x = \sigma E_x A$

$A = wd$

$\sigma = \rho \mu_n$

$$V_H = \frac{\sigma E_x B_z \mu_n wd}{\sigma w}$$

Using, $V_H = E_x B_z \mu_n d$

$V_H = 500 \times 10.5 \times 1.2 \times 5 \times 10^{-2}$

$V_H = 315$ Volts

Hence, the correct answer is 315.

Example 17:

Given that the Hall coefficient in a semiconductor sample is 2×10^{-3} m^3/C. Also, the resistivity is equal to 9×10^{-3} Ω-m. Find out the type of semiconductor and the mobility of charge carriers.

a) p-type, 4.5×10^{-3} m^2/V-sec

b) n-type, 0.02 m^2/V-sec

c) p-type, 0.22 m^2/V-sec

d) n-type, 0.45×10^{-3} m^2/V-sec

Sol:

Given, Hall coefficient (R_H) = 2×10^{-3} m^3/C

Resistivity = 9×10^{-3} Ω-m

Hall coefficient is positive. Hence, it is a p-type semiconductor.

$\mu \simeq \sigma R_H$

Using the formula of mobility:

$$\mu \simeq \frac{R_H}{\text{Resistivity}}$$

So, $\mu = \frac{2 \times 10^{-3}}{9 \times 10^{-3}} = 0.22\, m^2/\text{V-sec}$

Hence, the correct option is (C).

Chapter Summary

- The total energy of an electron is always negative.

$$W = -\frac{1}{8\pi \in_0 r} \cdot q^2$$

- A stationary state is determined by the condition that the angular momentum of an electron in this state is quantized and must be an integral multiple of $\frac{h}{2\pi}$.

Thus, $mvr = \frac{nh}{2\pi}$

- $E_g = \frac{1.24}{\lambda(\mu m)} eV$
- Classification of materials based on energy band diagram.

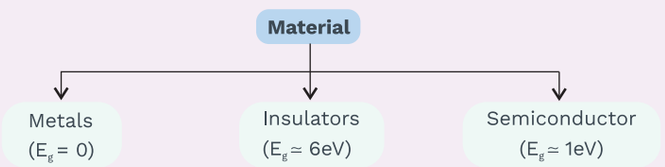


Fig. 1.45 Classification of Materials on the Basis of Energy Band Diagram

- $E_g(T) = E_{g_0} - \beta T$
- Direct band gap semiconductor has its valence band maxima and conduction band minima occur at the same value of k **(i.e. k = 0)**
- Indirect band gap semiconductor has its valence band maxima at different value of k than its conduction band minima.

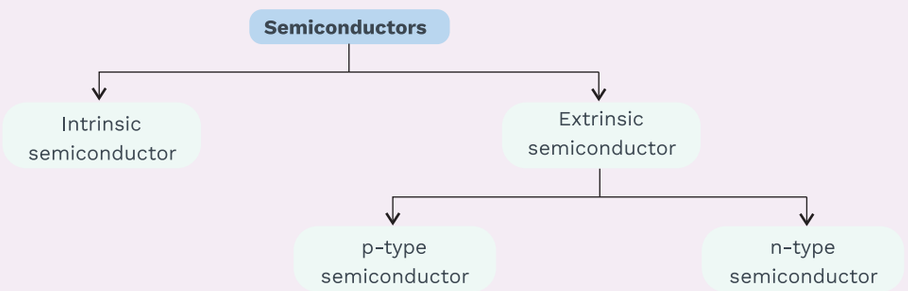


Fig. 1.46 Types of Semiconductor

- Holes are less mobile than electrons.
- In intrinsic semiconductor: $n = p = n_i$

- In n-type semiconductor:

$$n \simeq N_D$$

$$p = \frac{n_i^2}{N_D}$$

- In p-type semiconductor:

$$p \simeq N_A$$

$$n = \frac{n_i^2}{N_A}$$

- Mass action law: $np = n_i^2$
- Mass action law is not applicable to degenerative type semiconductor.
- Fermi energy is the highest energy value of electron at '0' K.
- In intrinsic semiconductor, Fermi level is present at the middle of energy band gap.
- In n-type, Fermi level is present above the midpoint of energy gap and below E_c. The exact location depends on doping level for a given temperature.
- In p-type, Fermi level is present below the midpoint of energy gap and above E_v.
- Degenerative semiconductors are those in which doping concentration is very high, and the Fermi level lies in the conduction band or valence band.
- Effect of temperature on Fermi level.

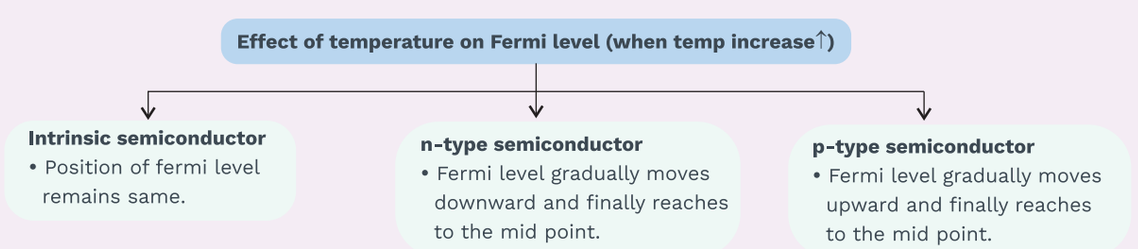


Fig. 1.47 Effect of Increase in Temperature on the Fermi Level

- Fermi-Dirac distribution

$$f(E) = \frac{1}{1 + e^{(E-E_F)/kT}}$$

- $N_c = 2\left(\frac{2\pi m_n^* kT}{h^2}\right)^{3/2}$
- $N_v = 2\left(\frac{2\pi m_p^* kT}{h^2}\right)^{3/2}$
- $F(E_c) = \frac{1}{1+e^{(E_c-E_F)/kT}} \simeq e^{-(E_c-E_F)/kT}$
- $n_0 = N_c\, e^{-(E_c-E_F)/kT}$
- $1-f(E_v) = \frac{1}{1+e^{-(E_v-E_F)/kT}} \simeq e^{-(E_F-E_v)/kT}$
- $p_0 = N_v\, e^{-(E_F-E_v)/kT}$
- $n_i = \sqrt{N_c N_v}\, e^{-E_g/2kT}$
- $n_i = \sqrt{A_0}\, T^{3/2} \exp\left[-E_g / 2kT\right]$
- Minority charge carrier concentration $= \frac{n_i^2}{\text{Majority charge carrier concentration}}$
- Minority charge carrier concentration × Doping concentration = n_i^2
- Position of Fermi level

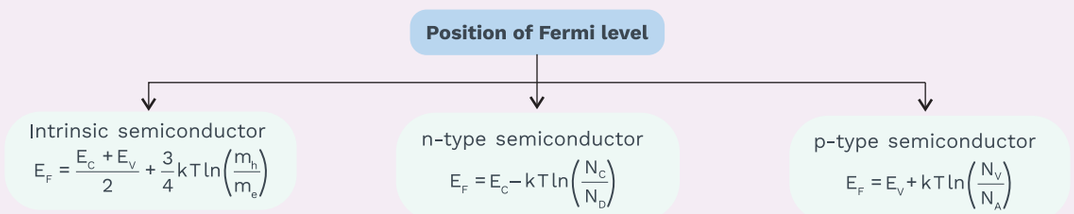


Fig. 1.48 Position of the Fermi Level in Different Types of Semiconductor

- Charge neutrality:
  Total positive charge = Total negative charge

  $N_D + p = N_A + n$
- For n-type semiconductor:

  $n_n = \frac{N_D}{2} + \sqrt{\left(\frac{N_D}{2}\right)^2 + n_i^2}$

- For p-type semiconductor:

$$p_p = \frac{N_A}{2} + \sqrt{\left(\frac{N_A}{2}\right)^2 + n_i^2}$$

- For compensated n-type semiconductor:

$$n = \frac{N_D - N_A}{2} + \sqrt{\left(\frac{N_D - N_A}{2}\right)^2 + n_i^2}$$

- For compensated p-type semiconductor:

$$p = \frac{N_A - N_D}{2} + \sqrt{\left(\frac{N_A - N_D}{2}\right)^2 + n_i^2}$$

- $E = -\frac{dV}{dx}$
- $v_{drift} = \frac{v_{max} + v_{min}}{2}$
- $v_d = \mu E$
- Effect of temperature on mobility:

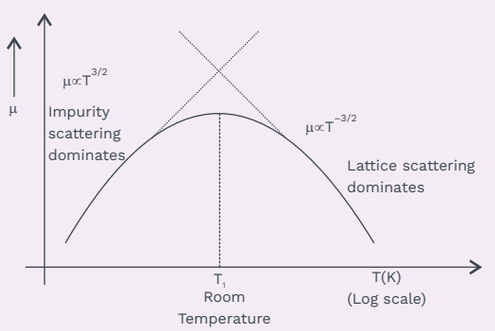


Fig. 1.49 Effect of Temperature on Mobility

- Effect of impurity dose on mobility:

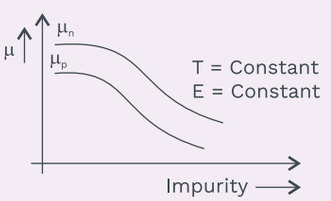


Fig. 1.50 Effect of Impurity Dose on Impurity

- Variation of mobility with electric field:

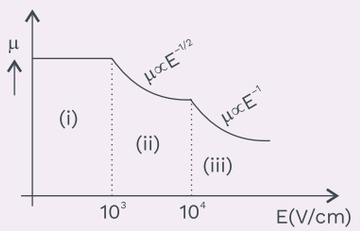


Fig. 1.51 Variation of Mobility eith Electric Field

- Variation of v_d with E:

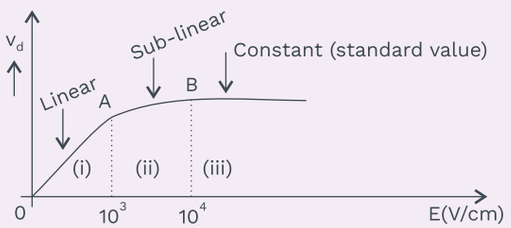


Fig. 1.52 Variation of Drift Velocity with Electric Field

- $v_{d_{max}} = 10^7$ cm / sec

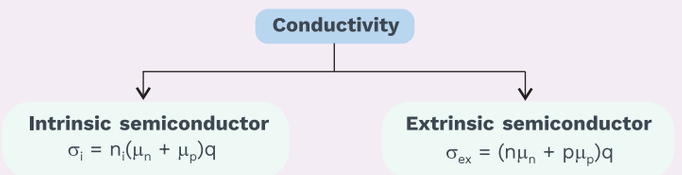


Fig. 1.53 Conductivity in Intrinsic and Extrinsic Semiconductor

- $J_{drift} = \sigma E$
- $J_{drift} = J_{n_{drift}} + J_{p_{drift}}$
- $J_{n(diff)} = qD_n \frac{dn(x)}{dx}$
- $J_{p(diff)} = -qD_p \frac{dP(x)}{dx}$
- $J_{diff} = J_{n(diff)} + J_{p(diff)}$
- $J_{diff} = qD_n \frac{dn(x)}{dx} - qD_p \frac{dp(x)}{dx}$
- $J_{total} = J_{drift} + J_{diff}$

- $L = \sqrt{D\tau}$
- Injection by illumination:

$$P(x) = P_{n_0} + P'_{n_0}\, e^{-x/L_p}$$

- Einstein relation:

$$\frac{D_n}{\mu_n} = \frac{D_p}{\mu_p} = \frac{kT}{q} = V_T$$

- Hall voltage $V_H = \frac{BI}{\rho w}$
- Hall coefficient $R_H = \frac{1}{\rho} = \frac{V_H w}{BI}$

$\because\ \rho = nq$

$\therefore R_H = \frac{1}{nq}$

- $\mu = \sigma R_H$
- If random thermal distribution is taken into account, then $\mu = \left(\frac{8\sigma}{3\pi}\right) \cdot R_H$

PRACTICE QUESTIONS

Q1 **If the longest wavelength that can be absorbed by a material is 1.12 μm, then the band gap of this material is:**

a) 1.21 eV
b) 0.854 eV
c) 1.107 eV
d) 0.710 eV

Q2 **In an intrinsic semiconductor material, calculate the factor by which the temperature has to be changed when intrinsic concentration is to be doubled while keeping the energy gap constant.**

a) $T_2 = 1.587\ T_1$
b) $T_2 = 1.381\ T_1$
c) $T_2 = 2T_1$
d) $T_2 = 1.414\ T_1$

Q3 **Consider the following statements. The intrinsic concentration of a semi -conductor:**

a) depends on doping
b) increases linearly with an increase in temperature
c) increases non-linearly with an increase in temperature
d) increases exponentially with the decrease of band gap of the semiconductor

Q4 **Consider G_e = at T = 300 K with donor concentration $N_D = 10^{15}/cm^3$ and $N_A = 0$. Calculate the position of the Fermi energy level with respect to the intrinsic Fermi level for this doping concentration.**

Q5 **Pure Ge has an electrical resistivity of 3500 Ω-m. If the free carrier density in it is $1.3 \times 10^6/m^3$, and the electron mobility is 2 times that of hole mobility, calculate the mobility values of electrons & holes.**

Q6 **A sample of silicon at temperature, T = 300 K of length 2.5 cm and area A = 2 mm^2 is doped with $10^{17}/cm^3$ of phosphorous (P) and $9 \times 10^{16}/cm^3$ of boron. The resistance of the sample is _____Ω. (Assume $\mu_p = 200\ cm^2/V\text{-}sec$, $\mu_n = 500\ cm^2/V\text{-}sec$).**

Q7 **The spectral response of certain elements at room temperature is shown below, then:**

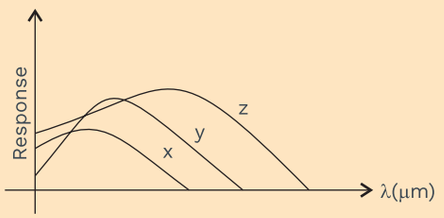


Fig. 1.54 Spectral Response

a) x = GaAs, y = Ge, z = Si **b) x = Si, y = GaAs, z = Ge**
c) x = GaAs, y = Si, z = Ge **d) x = Ge, y = Si, z = GaAs**

Q8 **The ratio of hole diffusion current to the electron diffusion current crossing the junction of p-n diodse if conductivity at p side is 1.5 J/cm and n side is 0.8 J/cm is ____. (Assume $L_n = L_p = 0.15$ cm)**

Q9 **Si is doped with boron (B), having a concentration of 4×10^{17} atoms/cm^3. The intrinsic carrier concentration of Si is 1.5×10^{10}/cm^3, and the value of KT/q is 25 mV at room temperature (300K). What is the Fermi level of doped silicon?**
a) Move up by 0.18 eV **b) Move down by 0.18 eV**
c) Move up by 0.427 eV **d) Move down by 0.427 eV**

Answer key:

1) c) **2) a)** **3) c, d)** **4) (0.095 eV)**

5) ($\mu_p = 4.57 \times 10^8$ m^2/V-sec, $\mu_n = 9.14 \times 10^8$ m^2/V-sec)

6) (156.25 Ω) **7) c)** **8) (1.88)** **9) d)**